



सेंद्रल ट्रान्समिशन यूटिलिटी ऑफ इंडिया लिमिटेड

(पावर ग्रिड कॉर्पोरेशन ऑफ इंडिया लिमिटेड के स्वामित्व में)

(भारत सरकार का उद्यम)

CENTRAL TRANSMISSION UTILITY OF INDIA LTD.

(A wholly owned subsidiary of Power Grid Corporation of India Limited)

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Date: 24-06-2024

As per distribution list

Subject: 30th Consultation Meeting for Evolving Transmission Schemes in Northern Region-Minutes of Meeting

Dear Sir/Ma'am,

Please find enclosed the minutes of the 30th Consultation Meeting for Evolving Transmission Schemes in Northern Region held on 18th June, 2024 (Tuesday) through virtual mode.

The minutes are also available at CTU website (www.ctuil.in)

Thanking you,

Yours faithfully,

(Handwritten signature: Kashish Bhambhani)
24/06/24

(Kashish Bhambhani)
General Manager (CTU)

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Minutes of 30th Consultation Meeting for Evolving Transmission Schemes in Northern Region held on 18.06.2024

The 30th Consultation Meeting for Evolving Transmission Schemes in the Northern Region (CMETS-NR) was held through VC on 18.06.2024. List of participants is enclosed at **Annexure-A**.

It was informed that the Minutes of the 29th CMETS-NR meeting held on 17/05/2024 were issued vide letter dated 21/05/2024. No comments were received on the minutes, accordingly, minutes were confirmed as circulated.

A. Network Expansion Scheme

A.1 Transmission system for evacuation of power from Rajasthan REZ Ph-V (Part-1 :4 GW) [Sirohi/Nagaur] Complex, Transmission scheme

It was deliberated that Renewable Energy Zones (REZs) were identified by MNRE/SECI with a total capacity of 181.5 GW for likely benefits by the year 2030 in eight states, which includes 75GW REZ potential in Rajasthan comprising of 15 GW Wind and 60 GW Solar. In this regard a Committee on Transmission Planning for RE was constituted by MOP for planning of the requisite Inter State Transmission System required for the targeted RE capacity by 2030 for which a Comprehensive transmission plan for evacuation of 75GW RE potential from Rajasthan was evolved.

Details of schemes approved/Under Planning scheme as part of above is as under:

S.No	Transmission Scheme	RE Potential	Status
A	Under Bidding/ Approved		
1	Rajasthan REZ Ph-IV (Part-1 :7.7GW) (Bikaner Complex)	14 GW (Solar 14GW, BESS:6GW) Bikaner-II : 3.7GW Bikaner-III: 4GW	Awarded
2	Rajasthan REZ Ph-IV (Part-2 :5.5GW) (Jaisalmer/Barmer Complex)	5.5GW (Solar) Fatehgarh-IV: 4 GW Barmer-I: 1.5 GW	Under Bidding
3	Rajasthan REZ Ph-IV (Part-3 :6GW) (Bikaner Complex)	6 GW (Solar) Bikaner-IV:6GW	Under Bidding

S.No	Transmission Scheme	RE Potential	Status
B	Planned/Under Planning		
1	Rajasthan REZ Ph-IV (Part-4 :3.5GW) (Jaisalmer/Barmer Complex)	3.5 GW (Solar) Fatehgarh-IV: 1 GW Barmer-I: 2.5 GW	Recently approved in NCT
2	Rajasthan REZ Ph-IV (Part-5 : 6GW) (Barmer Complex)	6 GW (Solar) Barmer-II : 6GW	Timeframe : 2029 (HVDC)
3	Rajasthan REZ Ph-IV (Part-6 : 6GW) (Bhadla/Bikaner Complex)	6 GW (Solar) Bhadla-IV: 2 GW Bikaner-V: 4 GW*	
4	Rajasthan REZ Ph-V (Part-1: 4GW) (Sirohi/Nagaur Complex)	4 GW (Solar) Sirohi: 2 GW Nagaur: 2 GW	Present proposal Timeframe : 2027 (EHVAC)

To evolve the transmission scheme, studies are carried out in two timeframes:

- 1) **Phase-1 (2027) for 4GW:** Transmission scheme for Rajasthan REZ Ph-V (Part-1: 4GW) (Sirohi/Nagaur Complex)
- 2) **Phase-2 (2029) for 12GW :** Transmission scheme for Rajasthan REZ Ph-IV (Part-5 : 6GW) (Barmer Complex) & Rajasthan REZ Ph-IV (Part-6 : 6GW) (Bhadla/Bikaner Complex)

It was stated that a Joint study meeting was held on 09.05.24 with stakeholders in NR to deliberate & finalize the Transmission system for evacuation of power from Rajasthan REZ Ph-V (Part-1 :4 GW) [Sirohi/Nagaur] Complex, Transmission scheme for Rajasthan REZ Ph-IV (Part-5 : 6GW) (Barmer Complex) & Rajasthan REZ Ph-IV (Part-6 : 6GW) (Bhadla/Bikaner Complex). As part of Rajasthan REZ Ph-V (Part-1 :4 GW) [Sirohi/Nagaur], following ISTS scheme was proposed in the Joint study meeting

Immediate Evacuation of Power from Sirohi PS (2GW)

- 5x500MVA, 400/220kV ICTs at Sirohi S/s along with transformer bays

Common Evacuation of Power from Sirohi PS (2GW) & Merta-II PS (2GW)

NR-WR Inter regional corridor

- Sirohi – Mandsaur PS 765KV D/c line (~ 320kms) along with 240 MVar switchable line reactor at Sirohi S/s end and 330 MVar switchable line reactor at Mandsaur PS end for each circuit of Sirohi – Mandsaur PS 765KV D/c line

Western region

- Mandsaur PS – Khandwa (New) 765kV D/c line (~230km.) along with 240MVar switchable line reactor for each circuit at each end of Mandsaur PS – Khandwa (New) 765kV D/c line

In the above Joint study meeting, Grid-India observations on proposed transmission schemes as well as on All India Study files (2027 & 2029 time frame) were deliberated and same was also communicated by Grid-India vide their mail dated 09.05.24 (Copy of Grid-India observations are enclosed in **Annexure-1**). Major observations on 2027 timeframe files are as under

- 1) LGB of All India file – Higher demand considered in NR,WR & ER region, Merit order of thermal generating units
- 2) Blocking of Reactive support from units with no generation
- 3) High loading and low voltages in the ER grid in solar peak hours
- 4) N-1 non compliance of ICTs in Rajasthan (Intra state and Inter state)
- 5) N-1 non compliance of ICTs in Western region
- 6) Critical loading in 400 kV Farakka – Kahalgaon D/C, 400 kV New Ranchi – PPSP D/C and ICTs in West Bengal in real time
- 7) Off loading of 765kV Ghiror – Aligarh D/c line
- 8) Critical loading and higher angular separation (>20 degree) under N-1 contingency in few EHVAC lines in NR as well as Low SCR of Sirohi PS
- 9) Very low voltages at various nodes in the system including in NR RE Complex
- 10) Provide Study cases for other scenarios (other than solar maximized scenario)

In the above Joint study meeting, it was decided that comprehensive schemes discussed may be segregated in two phases (Phase-1: Transmission scheme for Rajasthan REZ Ph-V (Part-1: 4GW) (Sirohi/Nagaur Complex) & Phase-2: Transmission scheme for Rajasthan REZ Ph-IV (Part-5 : 6GW) (Barmer Complex) & Rajasthan REZ Ph-IV (Part-6 : 6GW) (Bhadla/Bikaner Complex).

In Phase-1, PSS/E files shall be modified w.r.t discussion and comments received from Grid-India on 2027 time frame study files. Revised PSS/E files will be circulated to NR stakeholders for finalization of proposed scheme in next Joint study/CMETS-NR meeting

(Jun'24). After finalization of Phase-1 scheme i.e. Transmission scheme for Rajasthan REZ Ph-V (Part-1: 4GW) (Sirohi/Nagaur Complex), PSS/E files shall be modified for 2029 time frame w.r.t discussion and other observations/comments received from Grid-India on 2029 timeframe study files and subsequently proposal for Rajasthan REZ Ph-IV (Part-5 : 6GW) & REZ Ph-IV (Part-6 : 6GW)) scheme (Ph-2) will be taken up in Joint meeting/CMETS-NR meeting for stakeholders consultation.

Accordingly as part of Phase-1, revised PSS/E files for solar maximized scenario (SC-4 & 7) along with Scenario 8 (evening peak) & Scenario 9 (night off peak) files for 2027 time frame along with reply to Grid-India observations on 2027 timeframe files was sent on 04.06.24 to all stakeholders (Copy of CTU reply is enclosed in **Annexure-2**).

Subsequently, Grid India vide their mail dated 13.06.24 sent observations on revised PSS/E files (Copy of Grid-India observations enclosed in **Annexure-3**). In the CMETS-NR meeting, CEA agreed to the proposed transmission scheme. Grid-India stated that they do not have any specific comment on proposed transmission scheme, however their observation on other issues on 2027 timeframe files are as under:

- a) Higher demand considered in WR & ER region
- b) High loading in EHVAC lines in ER and in WR-ER in solar peak hours
- c) N-1 non compliance of ICTs in Rajasthan (Intra state and Inter state)
- d) Dynamic simulation studies incl. inertia studies for various contingency scenarios
- e) Critical loading and higher angular separation (>20 degree) under N-1/N-1-1 contingency of 765kV Bikaner-Moga D/c line as well as Low SCR of Sirohi PS

It was deliberated in CMETS-NR meeting that Transmission planning is carried out for peak load scenario. In view of that CTU analyze the historical data (Feb'23 to Feb'24) for solar maximized time and it is observed that load growth of more than 7 % is observed in winter solar peak hours in All India & most of the regions. Green Hydrogen and bulk consumer load is considered over and above to EPS demand in planning studies .

For high loading in EHVAC lines in ER, Grid-India stated that 400 kV Farakka – Kahalgaon D/c & 400 kV New Ranchi – PPSP D/c is overloaded in real time also. In view of RE addition in NR and WR and backing down of thermal units in ER, critical loading is observed in ER and WR-ER corridors in planning studies. In view of that strengthening is being taken up on priority with progressive addition of solar generation in next 2-3 years so that schemes in ER are available for absorption of RE power in Eastern region.

In the meeting, CTU stated that Reconductoring of some of the lines i.e. 400kV Farakka – Kahalgaon and 400kV Talcher – Meramundali is recently agreed in CMETS-ER meeting with a completion schedule of 18 months from allocation. In real time, loading of 400 kV New Ranchi – PPSP D/c is in order as line rating was considered lower, however in planning studies loadings are higher. Reconductoring of Ranchi New – New PPSP 400kV D/c line and any other augmentation in ISTS and commensurate STU network would be identified based on the joint study meeting scheduled next month with WBSETCL. Reconductoring of 400kV Farakka – Sagardighi and 400kV Rourkela – Chaibasa 400kV D/c line would be taken up for discussion in CMETS-ER meeting (Jul'24). An additional corridor from WR

to ER needs to be envisaged for relieve the loading around Ranchi area considering LGB of 2027 and 2029 timeframe and change in power flow pattern due to envisaged Green hydrogen load. Augmentation of ICTs in WBSETCL, DVC & Odisha area have been already informed to respective STUs in various CMETS-ER meetings and joint meetings. Matter will be further emphasized to STUs/DVC to take up augmentation of ICTs within time.

It was also decided that due to huge power flow in solar peak hours from NR-ER and WR-ER, and thermal backdown in ER, there may be other system strengthening requirement within ER incl. requirement discussed above. Such transmission strengthening requirements will be planned shortly and taken up for stakeholder consultation based on study results under various scenarios.

For intra state ICT overloading in Rajasthan, RVPN stated that for Hindaun ICT work awarded in Mar'24 (24 months schedule) and for Chittorgarh, Heerapura and Deedwana ICTs proposal is in process. CTU requested RVPN to expediate the approval of augmentation of ICTs so that same is available in 2027 timeframe.

CTU stated that Loading of 400/220kV Kota (PG) ICT is higher due to low dispatch of KTPS generation units. Loading of Bhinmal ICT is higher in Sc-7 due to increasing demand. Implementation of ICTs may take less time (18 months from approval) as compared to transmission schemes for RE evacuation (30 months from approval). It was stated that necessary augmentation (Kota and Bhinmal ICT) will be carried out in ISTS in consultation with stakeholders once space availability from TSP for Bhinmal and real time loading data from NRLDC is received for implementation in 2027 timeframe or earlier. NRLDC agreed for the same

For dynamic simulation studies and stability studies, CTU stated presently there are convergence issues being faced in dynamic simulation for 2027 scenario for which consultation with IIT Mumbai is also under progress. In view of that, Dynamic simulation studies for planning scenario may take some time.

In 30th CMETS-NR meeting, CTU requested that Grid-India may also share present time frame converged dynamic file which will help to resolve convergence issues in dynamics file for 2027 timeframe w.r.t Generator models (conventional/RE) used by Grid-India. In the CMETS-NR meeting Grid-India stated that they are also facing convergence issues for present timeframe file and prepared dynamics file with some assumptions. They also shared some dynamic data with CTU in the past but only dynamic data will not help to resolve convergence issue. However for preparation for planning stage dynamic file, CTU and Grid-India can work together to prepare load flow first w.r.t dynamics file and subsequent Dynamic data file (Dyr) shall be prepared with help of models available with Grid-India.

CTU also stated that considering the severe convergence issues in dynamics file, it will be converged on truncated network specific to RE pockets in Rajasthan rather than to converge the file on All India level. NRLDC stated that for transient studies, truncated network modelling will be sufficient.

Further, in the meeting CTU stated that to resolve the issue of higher angular separation (>20 degree) under N-1/N-1-1 contingency of 765kV Bikaner-Moga D/c line, suitable strengthening scheme is already under planning. As part of above scheme LILO of 765kV

Bikaner -Moga line is envisaged which will reduce the angular separation (<30 degree). The above scheme will be taken up in Jul'24 meeting.

Regarding observation for Inertia response of units, NRLDC stated that it may be checked in dynamics file once prepared for critical contingencies i.e. outage of large generation say 5-10GW in RE pockets. In the above condition, inertial response of generation units, Grid frequency variation, df/dt, synchronous condenser response may be analyzed. Same was agreed.

For low SCR at Sirohi PS(4.2), CTU stated that no additional connectivity will be granted beyond 2 GW at 220kV level of Sirohi PS with proposed system (400/220kV ICTs at Sirohi PS+765kV Sirohi - Mandsaur). With growing interconnection and planned network in future, SCR at Sirohi may increase

Copy of CTU reply as well as gist of detailed deliberations held in present CMETS-NR meeting on all other Grid-India observation is enclosed in **Annexure-4**. Result of system studies (after minor modifications suggested by Grid-India in PSS/E file as well as deliberation held in CMETS-NR meeting) enclosed in **Exhibit-I**

Grid-India also enquired about space for STATCOM at Sirohi PS. CTU stated that 765/400kV Sirohi S/s are under bidding and to be implemented as part of Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part 2: 5.5 GW) (Jaisalmer/Barmer Complex).CTU stated that provision of space was kept at Sirohi PS for installation of STATCOM in future based on requirement.

Further, it was stated that CTUIL has received applications of connectivity of 5150 MW from Pumped Storage Projects from Greenko & Avaada near Robertganj in Sonbhadra District Uttar Pradesh. In view of the above, proposed ISTS transmission system for evacuation of power from PSP projects is planned and incorporated in 2027 timeframe files. Details of broad scheme is as under:

- Establishment of 5x1500 MVA 765/400 kV Robertsganj PS near Robertsganj area in Sonbhadra distt. (Uttar Pradesh) along with 2x240 MVA 765 kV & 2x125 MVA 400 kV bus reactors
- LILO of both circuits of 765 kV Varanasi- Gaya 2xS/c line at Robertsganj PS(~50 km)
- Establishment of 765kV Prayagraj S/s near Prayagraj (UP) along with 2x240 MVA 765 kV & 2x125 MVA 400 kV bus reactors
- LILO of 765 kV Fatehpur-Varanasi S/c line at Prayagraj PS(~30 km)
- LILO of 765 kV Fatehpur-Sasaram S/c line at Prayagraj PS(~30 km)
- Robertsganj PS – Prayagraj S/s 765 kV D/c line (~200 km) along with 240 MVA line reactor at both ends of each circuit of Robertsganj PS – Prayagraj S/s 765 kV D/c line

Additionally to resolve critical loading of 765kV Vindhyachal-Varanasi D/c line as well as for evacuation of PSP projects in UP (as indicated above), following ISTS scheme is proposed and incorporated in PSS/E files :

- 765kV Vindhyachal Pool - Prayagraj D/c line (about 218kms) along with 240MVA line reactor (switchable) at Prayagraj end on each ckt of 765kV Vindhyachal Pool - Prayagraj D/c line

- Bypassing of both ckts of 765kV Sasan – Vindhyachal Pool 2xS/c line at Vindhyachal Pool and connecting it with 765kV Vindhyachal Pool - Prayagraj D/c line, thus forming 765kV Sasan - Prayagraj D/c line

The above transmission schemes i.e. transmission scheme for evacuation of power from PSP plants as well as transmission scheme to resolve critical loading of 765kV Vindhyachal-Varanasi D/c line will be taken up in separate joint study meeting/ next CMETS-NR meeting for stakeholder consultation.

No other comments were received from Stakeholders on proposed scheme. Considering receipt of connectivity application of new RE generators in Sirohi & Nagaur/Merta complexes & based on discussion, following connectivity transmission scheme is agreed as part of Rajasthan REZ Ph-V (Part-1 :4 GW) scheme:

1. Transmission system for immediate Evacuation of Power from Sirohi S/s (2GW)

- 5x500MVA, 400/220kV ICTs at Sirohi S/s along with associated transformer bays
- 6 nos. 220kV line bays at Sirohi S/s for RE interconnection
- 220 kV Sectionalizer bay (1 set) along with 220 kV BC (2 Nos.) bay and 220 kV TBC (2 Nos.) bay at Sirohi S/s

2. Transmission system for immediate Evacuation of Power from Merta-II S/s (2GW*)

- 3x500MVA, 400/220kV ICTs at Merta-II S/s along with associated transformer bays
- 4 nos. 220kV line bays at Merta-II S/s for RE interconnection
- 220 kV Sectionalizer bay (1 set) along with 220 kV BC (1 No.) bay and 220 kV TBC (1 No.) bay at Merta-II S/s

3. Transmission system for Common Evacuation of Power from Sirohi PS (2GW) & Merta-II PS (2GW*)

a) NR-WR Inter regional corridor

- Sirohi – Mandsaur PS 765KV D/c line (~ 320kms) along with 240 MVar switchable line reactor at Sirohi S/s end 330MVar switchable line reactor at Mandsaur PS end for each circuit of Sirohi – Mandsaur PS 765KV D/c line

b) Western region

- Mandsaur PS – Khandwa (New) 765kV D/c line (~230km.) along with 240MVar switchable line reactor for each circuit at each end of Mandsaur PS – Khandwa (New) 765kV D/c line

***Transmission scheme for immediate evacuation of 1GW RE from Merta-II S/s to facilitate connectivity to RE generation developers was deliberated and approved in 19th NCT meeting held on 29.04.24**

A.2 Revision of Transmission system for evacuation of power from Luhri Stage-I HEP

It was stated that Transmission system for evacuation of power from Luhri Stage-I HEP was discussed and agreed in the 8th NCT meeting held on 25.03.2022. Subsequently, in the 14th NCT meeting held on 09.06.2023, it was decided that the timeframe of the above transmission scheme would be revised to 31st August 2026. The above transmission scheme was notified in Gazette dated 02.06.2022 and RECPDCL was appointed as the BPC of the transmission scheme. The transmission scheme is currently under bidding.

During the course of bidding, the logistics issues in transportation of large size equipment was highlighted by bidders. Regarding this matter, multiple meetings were held.

Recently, in the meeting taken by Addl. Secretary (Trans), MoP on 07.06.2024 to discuss the Logistics issues in the ongoing bidding of Luhri Stage -I HEP Transmission scheme, SJVN suggested to shift the location of Nange (ISTS) Pooling Station (presently proposed at Ogli village) to a suitable location near Koldam area adjacent to National Highway (NH) in order to remove the hurdles associated with transportation of heavy equipment to the proposed location of pooling station and accordingly, SJVN would bring 220 kV dedicated line from Luhri-I & Sunni Dam to the new ISTS Pooling station near Koldam. Further, SJVN has informed that the new location of ISTS Pooling station would be about 6-7 kms (BEE length) from Koldam HEP switchyard as per initial survey. After deliberations, it was decided to include SJVN proposal as an agenda item in the 20th NCT meeting.

CEA vide mail dated 13.06.24 to CTUIL requested to review the transmission scheme considering the change/modification proposed by SJVN. Further, details of modified scheme along with cost estimates may kindly be furnished by 18.06.2024 as the scheme has to be deliberated in the NCT meeting on 21.06.2024.

In view of revised location of Pooling station (near Koldam), CTU vide mail dated 14.06.2024 to all stakeholders informed that the new pooling station is now proposed to be established near Koldam (about 6-7 km) instead of Nange. Due to this, the length of the 400 kV New Pooling Station-Ropar line has reduced to about 70 km (from earlier expected 120 km (approx)). This has led to the overcompensation of 400 kV New Pooling Station-Ropar line (~95%) considering 50 MVAR line reactor at Ropar end. In view of the above, CTU in their mail (14.06.24) sought stakeholders views on deletion of 50 MVAR line reactor at Ropar S/s due to reduced line length. (70 km), however, views were awaited therefore discussion was decided to be taken up in this CMETS-NR meeting.

In the meeting, PSTCL stated that due to reduction in line length, line reactor deletion at Ropar end may be required, however they will re-confirm the decision with management discussion. CTU stated that proposal is to be discussed in next NCT meeting for which they need to provide inputs to CEA by today evening. In view of that till management decision received from PSTCL end, PSTCL confirmation for deletion of line reactor in this CMETS-NR meeting shall prevail. CEA and Grid-India also agreed for deletion of 50 MVAR line reactor at Ropar end.

CTU enquired PSTCL for any provision of bus reactor at 400/220kV Ropar S/s. PSTCL stated that they have not yet planned any bus reactor at Ropar S/s due to 50MVAR proposed line reactor at Ropar end. CTU stated that PSTCL may plan bus reactors (preferably

125 MVar) on all 400kV existing and planned STU substations incl. Ropar S/s subject to space availability as part of intra state schemes for better voltage regulation specially in winter night off peak scenario.

After deliberations, deletion of 50 MVar line reactor at Ropar due to over compensation(~95%) was agreed in transmission scheme for evacuation of power from Luhri Stage-I HEP.

Accordingly, the proposed revised scheme for evacuation of power from Luhri Stage-I is as below:

Sl. No.	Earlier Scope of Transmission Scheme	Revised Scope of Transmission Scheme
1	Establishment of 7x105 MVA, 400/220kV Nange GIS Pooling Station along with 125 MVAR (420kV) Bus Reactor at Nange (GIS) PS(1-Ph units along with one spare unit) • 315MVA, 400/220kV ICT: 2 nos.(7x105 MVA including 1 spare ICT) • 400kV ICT bays: 2 nos. • 220kV ICT bays: 2 nos. • 400 kV, 125 MVar Bus Reactor # – 1 no. • 400 kV Bus Reactor bay- 1 no. • 400 kV Line Bays- 2 nos. Future provisions: Space for • 400/220kV ICTs (315 MVA with single phase units) along with associated bays: 3 nos. • 400 kV line bays along with switchable line reactor: 3 nos. • 220 kV line bays: 10 nos. • 220kV bus sectionalizer: 1 set	Establishment of 7x105 MVA, 400/220kV Pooling Station near Koldam (GIS) along with 125 MVAR (420kV) Bus Reactor (1-Ph units along with one spare unit) • 315MVA, 400/220kV ICT: 2 nos. (7x105 MVA including 1 spare ICT) • 400kV ICT bays: 2 nos. • 220kV ICT bays: 2 nos. • 400 kV, 125 MVar Bus Reactor # – 1 no. • 400 kV Bus Reactor bay- 1 no. • 400 kV Line Bays- 2 nos. Future provisions: Space for • 400/220kV ICTs (315 MVA with single phase units) along with associated bays: 3 nos. • 400 kV line bays along with switchable line reactor: 3 nos. • 220 kV line bays: 10 nos. • 220kV bus sectionalizer: 1 set
2	Nange (GIS) Pooling Station – Koldam 400 kV D/C line (Triple snowbird) (only one circuit is to be terminated at Koldam while second circuit would be connected to bypassed circuit of Koldam – Ropar/Ludhiana 400kV D/C line)-40 km	Pooling Station near Koldam (GIS)– Koldam (NTPC) 400 kV D/C line (Triple snowbird) (only one circuit is to be terminated at Koldam(NTPC) while second circuit would be connected to bypassed circuit of Koldam(NTPC) – Ropar/Ludhiana 400kV D/C line) – 7 km
3	1 no. of 400kV line bay at Koldam S/S for termination of Nange (GIS) Pooling Station – Koldam 400 kV line along with 125 MVAR	1 no. of 400kV line bay at Koldam S/s for termination of Pooling Station near Koldam (GIS)– Koldam(NTPC) 400 kV

Sl. No.	Earlier Scope of Transmission Scheme	Revised Scope of Transmission Scheme
	(420kV) Bus Reactor at Koldam S/s (1-Ph units along with one spare unit) 400 kV Line Bay- 1 no. 400 kV, 125 MVAR Bus Reactor# - 1 no. 400 kV, 125 MVAR Bus Reactor# - 1 no. (to be terminated in existing line bay at Koldam, which would be available due to bypassing of one circuit of Koldam – Ropar/Ludhiana 400 kV D/c line at Koldam S/s)	line along with 125 MVAR (420kV) Bus Reactor at Koldam(NTPC) S/s (1-Ph units along with one spare unit) 400 kV Line Bay- 1 no. 400 kV, 125 MVAR Bus Reactor# - 1 no. (to be terminated in existing line bay at Koldam(NTPC), which would be available due to bypassing of one circuit of Koldam – Ropar/Ludhiana 400 kV D/c line at Koldam(NTPC) S/s)
4	Bypassing one ckt of Koldam – Ropar/Ludhiana 400kV D/C line (Triple snowbird) at Koldam and connecting it with one of the circuit of Nange-Koldam 400kV D/C line (Triple snowbird), thus forming Nange- Ropar/ Ludhiana one line (Triple snowbird)	Bypassing one ckt of Koldam(NTPC) – Ropar/Ludhiana 400kV D/C line (Triple snowbird) at Koldam(NTPC) and connecting it with one of the circuit of Pooling Station near Koldam (GIS) – Koldam(NTPC) 400kV D/c line (Triple snowbird), thus forming Pooling Station near Koldam – Ropar/ Ludhiana one line (Triple snowbird)
5	1x50 MVAR switchable line reactor at Ropar end of Nange-Ropar/Ludhiana 400kV line 400 kV, 50MVAR Line Reactor- 1 no. 400 kV Reactor Bay- 1 no	--- TO BE DELETED --- Deleted as the line length is reduced to about 70 km resulting into reactive overcompensation (~95%).

Note:

- (i) NTPC to provide space for 400 kV line bays and Bus Reactor at Koldam(NTPC) s/s.
- (ii) The line lengths mentioned above are approximate as the exact length shall be obtained after the detailed survey.
- (iii) #125MVAR bus reactor at New Pooling Station and Koldam(NTPC) s/s shall consist of 3x41.67 MVAR single phase units and 1x41.67MVAR as spare unit.

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Annexure-A**List of Participants of 30th Consultation meeting for Evolving Transmission Schemes in NR held on 18.06.2024****CEA**

Shri Nitin Deswal	Deputy Director
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Grid India

Shri Priyam Jain	Ch. Manager
Shri Gaurav Malviya	Manager
Shri Raj Kishan	Asst. Manager

SECI

Shri R.K.Agarwal	Consultant
------------------	------------

CTU

Shri V Thiagarajan	Sr. GM (CTU)
Shri Kashish Bhambhani	GM (CTUIL)
Shri Sandeep Kumawat	DGM (CTU)
Shri R Narendra Sathvik	Ch. Manager (CTU)
Shri Pratyush Singh	Ch. Manager (CTU)
Shri Yatin Sharma	Manager (CTU)
Shri Anupam Kumar	Manager (CTU)
Shri Madhusudan Meena	Engineer (CTU)

Powergrid

Shri Rohit Kumar Jain DGM

HVPNL

Shri Sanjay Verma SE Planning

RVPNL

Shri Dr. Om Prakash Mahela XEN(PP&D)

PSTCL

Smt. Ranjana AEN

UPPTCL

Shri. Satendra Kumar SE

PTCUL

Shri. Ashok Kumar EE

Shri Lalit Kumar SE

NLDC Inputs on Rajasthan REZ Phase-IV and V Schemes

1. Schemes proposed for:

- a) Rajasthan REZ Ph-IV (Part-5 : 6GW) (Barmer Complex)
- b) Rajasthan REZ Ph-IV (Part-6 : 6GW) (Bhadla/Bikaner Complex)
- c) Rajasthan REZ Ph-V (Part-1: 4GW) (Sirohi/Nagaur Complex)

S.No	Transmission Scheme	RE Potential	Status
A Under Bidding/ Approved			
1	Rajasthan REZ Ph-IV (Part-1 :7.7GW) (Bikaner Complex)	14 GW (Solar 14GW, BESS:6GW) Bikaner-II : 3.7GW Bikaner-III: 4GW	Awarded
2	Rajasthan REZ Ph-IV (Part-2 :5.5GW) (Jaisalmer/Barmer Complex)	5.5GW (Solar) Fatehgarh-IV: 4 GW Barmer-I: 1.5 GW	Under Bidding
3	Rajasthan REZ Ph-IV (Part-3 :6GW) (Bikaner Complex)	6 GW (Solar) Bikaner-IV:6GW	Under Bidding (Recently approved in NCT)
B Planned/Under Planning			
1	Rajasthan REZ Ph-IV (Part-4 :3.5GW) (Jaisalmer/Barmer Complex)	3.5 GW (Solar) Fatehgarh-IV: 1 GW Barmer-I: 2.5 GW	Recently approved in NCT
2	Rajasthan REZ Ph-IV (Part-5 : 6GW) (Barmer Complex)	6 GW (Solar) Barmer-II : 6GW	Timeframe : 2029 (HVDC)
3	Rajasthan REZ Ph-IV (Part-6 : 6GW) (Bhadla/Bikaner Complex)	6 GW (Solar) Bhadla-IV: 2 GW Bikaner-V: 4 GW*	
4	Rajasthan REZ Ph-V (Part-1: 4GW) (Sirohi/Nagaur Complex)	4 GW (Solar) Sirohi: 2 GW Nagaur: 2 GW	
			Present proposal Timeframe : 2027 (EHVAC)

NLDC Inputs on Rajasthan REZ Phase-IV and V Schemes

2. Inputs on AC Scheme proposed for Rajasthan REZ Ph-V (Part-1: 4GW) (Sirohi/Nagaur Complex)

Proposed System (Timeframe – 2027):

- 5x500MVA, 400/220kV ICTs at Sirohi S/s along with transformer bays
- Sirohi – Mandsaur PS 765KV D/c line (~ 320 kms) along with 330 MVAR switchable line reactor for each circuit at each end of Sirohi – Mandsaur PS 765KV D/c line
- Mandsaur PS – Khandwa (New) 765kV D/c line (~230km.) along with 240MVAR switchable line reactor for each circuit at each end of Mandsaur PS – Khandwa (New) 765kV D/c line

a) Load-Generation Balance and IR Flows

S. No.	Region	Load (MW)	Generation (MW)	IR Exchange (MW)	20th EPS Peak Load in 2026-27 (MW)	Average Demand Met in Solar Hours in Feb 2024 (MW)
1	NR	78239	109488	31249	97898	59664
2	WR	98877	92846	-6030	89547	72184
3	SR	78984	94860	15876	80864	62000
4	ER	29496	14828	-14668	37265	20000
5	NER	2941	2654	-287	4855	2000
6	All India	288536	314676	x	277201	216185

S. No.	State	Load (MW)	Generation (MW)	IR Exchange (MW)	20th EPS Peak Load in 2026-27 (MW)	Average Demand Met in Solar Hours in Feb 2024 (MW)
1	Rajasthan	19376	81119	31249	21175	17000

- i. Demand considered at all India level as well as in some of the regions in Feb 2027 period is even more than the peak demand of the year 2026-27 as per 20th EPS report.
- ii. The actual average demand met during Feb'24 solar hours has also been provided for reference. It appears that the growth demand considered in NR, WR, ER, and at all India level is on the higher side. Same may be reviewed.

iii. Solar and Thermal Generation

- ~190000 MW solar generation considered in service at all India level
- ~62000 MW solar generation is in service in Rajasthan

NLDC Inputs on Rajasthan REZ Phase-IV and V Schemes

- ~22000 MW RE generation in Khavda complex has been considered in the case
 - Thermal generation has been uniformly backed down to 55% (especially in NR and WR) at most of the stations. It is suggested that the generation may be scaled as per the merit order. Cheaper thermal generating stations like Singrauli and Rihand are also running at 55% level.
 - Some of the thermal generating stations (like Vadinar, Khaparkheda, Tiroda-APL etc.) are running below 40% tech min level also. Same may be clarified.
 - Many units with no active power generation ($P_{gen} = 0$) are kept on in the case. These units are also providing reactive power support in the study case. The voltages in the case, therefore, may not represent the actual scenario. Such units shall therefore be kept off in the case.
- iv. With above-mentioned corrections in demand and generation, the whole LGB of the study case will change. This might result in significant changes in network flows also. Therefore, it is suggested that the proposed system may be tested on realistic load-generation scenario for 2026-27, preferably the one coming out of production cost modelling studies.
- v. There is significant amount of power flow towards eastern region during solar hours resulting in very high loading and low voltages in the ER grid. It is suggested that system augmentation in ER may be taken up on priority so that the required system gets commissioned with the associated RE generation (2026-27 timeframe).

The details of N-1 non-compliance, high line loading, and low voltages are provided in subsequent sections.

b) Network Loadings

- i. With the proposed scheme, there is improvement in loading of 400 kV RAPP – Shujalpur D/C.
- ii. Several ICTs in Rajasthan are loaded much above the N-1 limit (Deedwana - PG, Hindaun, Hirapura, Chittorgarh, Kota – PG etc.)
- iii. In the base case, several ICTs in WR are violating N-1 criteria (765/400 kV Jamnagar - ISTS, 765/400 kV Indore – PG, 400/220 kV Mandsaur, 400/220 kV Hazira, 400/132 kV Kirnapur – MP, 400/220 kV Navi Mumbai etc.)
- iv. Critical loading in 400 kV Farakka – Kahalgaon D/C, 400 kV New Ranchi – PPSP D/C and ICTs in West Bengal is already observed in real-time during solar hours. These elements are severely loaded in the base case also (2027 timeframe).

NLDC Inputs on Rajasthan REZ Phase-IV and V Schemes

Details of Lines and ICTs loaded around/more than the N-1 loading limit in NR, WR & ER are given in **Annexure – 1**.

- v. 765 kV Ghiror – Aligarh is completely off-loaded (5 MW loading in each ckt)
- vi. 765 kV Aligarh – G.Noida touches 2800 MW (Scenario – 7) & 3400 MW (Scenario – 4) on N-1 of 765 kV Aligarh – Kanpur
- vii. Angular difference between 765 kV Bikaner and 765 kV Moga touches 26° under N-1
- viii. RATE-2 limits of several lines are not available. The same may be corrected in the cases shared by CTUIL.
- ix. Angular Separation:

Scenario	765 kV Sirohi – Mandsaur D/C Line	765 kV Sirohi – Risabhdeo D/C	765 kV Neemuch – Khandwa D/C
Base Case	14.4° (1746 MW)	9.5° (1744 MW)	7.3° (1230 MW)
Under N-1 Contingency	17.5° (2100 MW)	12.4° (2254 MW)	9.3° (1581 MW)
Under N-1-1 Contingency	22.3°	17.9°	13.2°

c) Voltages

- i. Very low voltages have been experienced at various nodes in the system including in NR RE Complex.
- ii. The voltages under N-1 condition (depleted network conditions) will further dip.

765 kV Level

Name	Base kV	Bus No.	Voltage (p.u.)	Voltage (kV)
SIWANI	765	147880	0.94	719
NORTH CHN	765	548008	0.95	730
NCTPS-STG3	765	548013	0.95	730
MADHUGIRI	765	528001	0.95	730
ETPSREP7	765	548124	0.95	730
TIRUVLM	765	548087	0.95	730
ARIYALUR7	765	548127	0.95	730
BIKANER-IV	765	167000	0.96	736
MOGA-PG	765	137703	0.96	738
SIKAR	765	167497	0.97	740
BHADLA-2	765	167498	0.97	743

NLDC Inputs on Rajasthan REZ Phase-IV and V Schemes

Name	Base kV	Bus No.	Voltage (p.u.)	Voltage (kV)
BIKANER-NW	765	167458	0.97	745
NARELA	765	157001	0.97	745
KHETRI	765	167774	0.97	746
MUL	765	358129	0.97	746
JHATI-PG	765	157708	0.98	746
KURAWAR-7	765	368086	0.98	747
KURN4_7	765	508887	0.98	747
KURNOOL-III	765	508049	0.98	748
MAHESHWARAM	765	518051	0.98	749
BHIWN-PG	765	147704	0.98	749

400 kV Level

Name	Base kV	Bus No.	Voltage (p.u.)	Voltage (kV)
YELAHNKA	400	524084	0.93	373
DHANLLI	400	524025	0.93	374
JEY-BOL_FSC	400	424005	0.94	374
HDURGA	400	524102	0.94	374
PEENYA4	400	524169	0.94	374
MYSORE4	400	524010	0.94	374
NELMANG4	400	524007	0.94	375
BIDADI	400	524077	0.94	375
PULNTOPE4	400	544125	0.94	375
MANALI4	400	544040	0.94	376
KORATUR4	400	544035	0.94	376
THERVOI4	400	544126	0.94	376
HOODI4	400	524005	0.94	377
SMNH	400	524001	0.94	377
TARAMANI4	400	544015	0.94	377
MYLSNDRA	400	524035	0.94	377
MADHUGI4	400	524098	0.94	377
KADAKOLA	400	524106	0.94	377
GUINDY4	400	544133	0.94	378
KOYAMBEDU4	400	544019	0.95	378
SHOLNGNR	400	544091	0.95	378
MUL	400	354129	0.95	378
FSC-4	400	504999	0.95	379
FSC-3	400	504998	0.95	379
SIWANI	400	144880	0.95	379
DOMSANDRA	400	524171	0.95	379
MALEKTT	400	544086	0.95	379
SVCHTRM	400	544090	0.95	379
SPBUDUR4	400	544002	0.95	379
ALMATI4	400	544022	0.95	380
CHENN-EX	400	544000	0.95	380

NLDC Inputs on Rajasthan REZ Phase-IV and V Schemes

d) Short Circuit Ratio

Bus/Station	Connected RE Capacity	Fault Level	SCR (Without Current Contribution of immediate RE to be Connected)
220 kV Sirohi	2000 MW	8418 MVA	4.2
220 kV Merta – II	1000 MW	5734 MVA	5.7
400 kV Merta - II	1000 MW	20226 MVA	20.2

- e) **Dynamic Simulation Studies** – With more than 62,000 MW solar generation and three HVDC terminals in Rajasthan in close vicinity, it is important that stability aspects are also studied in detail at the planning stage. The generation is also getting evacuated through large EHV lines and hence, transient stability analysis also becomes important. Therefore, it is suggested that the results of the dynamic simulation studies may be shared.
- f) **Study cases for other scenarios** – The study cases for evening and solar peak scenario (Feb and June) have been shared. The study cases for balance 06 scenarios especially the off-peak cases may also be shared so that voltage related issues may also be examined in detail.

NLDC Inputs on Rajasthan REZ Phase-IV and V Schemes

3. Inputs on proposed Transmission Schemes for:

3.1: Rajasthan REZ Ph-IV (Part-5: 6GW) (Barmer Complex)

3.2: Rajasthan REZ Ph-IV (Part-6: 6GW) (Bhadla/Bikaner Complex)

Proposed Transmission scheme for Rajasthan REZ Ph-IV (Part-5: 6GW) (Barmer Complex) - (Timeframe – 2029)

- Establishment of 400/220kV, 6x500MVA S/s at suitable location near Barmer (Barmer-II Substation) along with 2x125 MVAR bus reactor
- LILO of both ckts of 400kV Fatehgarh-IV PS - Barmer-I PS at Barmer-II PS
- Establishment of 6000 MW, $\pm$ 800 kV Barmer-II (HVDC) [LCC] terminal station (4x1500 MW) at a suitable location near Barmer-II substation
- Establishment of 6000 MW, $\pm$ 800 kV South Kalamb S/s (HVDC) [LCC] terminal station (4x1500 MW) at a suitable location near South of Kalamb
- Establishment 2x1500MVA, 765/400kV Substation near South of Kalamb with 2x330 MVAR, 765 kV bus reactor and 2x125 MVAR, 420 kV bus reactor
- LILO of Pune-III – Boisar-II 765kV D/c line at South Kalamb with associated bays at South Kalamb S/s
- $\pm$ 800 kV HVDC line between Barmer-II (HVDC) & South Kalamb (HVDC) (with Dedicated Metallic Return) (1000kms)

Proposed Transmission scheme for Rajasthan REZ Ph-IV (Part-6: 6GW) (Bhadla/Bikaner Complex) - (Timeframe – 2029)

- Establishment of 765/400kV, 4x1500 MVA S/s & 400/220kV, 2x500MVA pooling station at suitable location near Bikaner (Bikaner-V PS)
- LILO of both ckts of 400kV Bikaner-II PS- Khetri D/c line at Bikaner-V PS (20km)
- Establishment of 765/400kV, 3x1500 MVA S/s & 400/220kV 3x500MVA pooling station at suitable location near Bhadla (Bhadla-IV PS) along with 2x125 MVAR & 2x240 MVAR bus reactor
- Bhadla-IV PS – Bikaner-V 765KV D/c line (~ 150kms) along with 240 MVAR switchable line reactor
- for each circuit at Bhadla-IV PS end of Bhadla-IV PS – Bikaner-V PS 765kV D/c line
- Bhadla-IV PS – Bhadla-III PS 400kV D/c line (Quad) (~30kms)

NLDC Inputs on Rajasthan REZ Phase-IV and V Schemes

- Establishment of 6000 MW, ± 800 kV Bikaner-V (HVDC) [LCC] terminal station (4x1500 MW) at suitable location near Bikaner
- Establishment of 6000 MW, ± 800 kV Begunia (HVDC) [LCC] terminal station (4x1500 MW) at
- Begunia (Distt. Khordha), Orissa
- ± 800 kV HVDC line between Bikaner-V (HVDC) & Begunia (HVDC) (with Dedicated Metallic
- Return) - 1900 km
- Establishment of 765/400kV, 5x1500 MVA S/s substation station at Begunia along with 2x125
- MVar & 2x240 MVar bus reactor
- Begunia - Paradeep (ISTS)* 765kV D/c line along with associated bays at both ends
- Begunia – Gopalpur (ISTS)# 765kV D/c line along with associated bays at both ends (150km)
- along with 240MVar switchable line reactor for each circuit at Begunia end of Begunia –Gopalpur
- (ISTS) 765kV D/c line
- Begunia – Khuntuni (OPTCL)@ 765kV D/c line

a) Load-Generation Balance

S. No.	Region	Load	Generation	IR Exchange	20th EPS Peak Load in 2028-29	Average Demand Met in Solar Hours in Feb 2024
1	NR	85756	129923	44167	109714	59664
2	WR	109083	107587	-1496	100246	72184
3	SR	81317	103944	22627	91285	62000
4	ER	43981	19628	-24352	42546	20000
5	NER	2869	2655	-214	5481	2000
6	All India	323005	363737	x	313098	216185

S. No.	State	Load	Generation	IR Exchange	20th EPS Peak Load in 2028-29	Average Demand Met in Solar Hours in Feb 2024
1	Rajasthan	21377	96462	75086	23590	17000
2	Odisha	21379	8599	-12780	8514	4300

NLDC Inputs on Rajasthan REZ Phase-IV and V Schemes

- i. Demand considered at all India level as well as in some of the regions in Feb 2029 period is even more than the peak demand of the year 2028-29 as per 20th EPS report.
- ii. The actual average demand met during Feb'24 solar hours has also been provided for reference. It appears that the growth demand considered in WR, ER, and at all India level is on the higher side. Same may be reviewed.
- iii. In the 20th EPS report, following is also mentioned:

*“Factors like reduction in transmission & distribution losses, energy efficiency improvement measures, **production of green hydrogen, penetration of electric vehicles**, roof-top solar, solar pumps etc. have appropriately been factored in the electricity demand forecast.”*

Therefore, the increase in demand due to green hydrogen load has already been captured in the 20th EPS figures.

iv. Solar and Thermal Generation

- ~225408 MW solar generation considered in service at all India level
 - ~89510 MW solar generation is in service in Rajasthan
 - ~42000 MW solar generation in Gujarat and ~24500 MW Khavda complex has been considered in the case
 - Thermal generation has been uniformly (especially in NR and WR) backed down to 55% at most of the stations. It is suggested that the generation may be scaled as per the merit order. Cheaper thermal generating stations like Singrauli and Rihand are also running at 55% level.
 - Some of the thermal generating stations (like Chandrapur, Vadinar, Khaparkheda, Tiroda-APL etc.) are running below 40% tech min level also. Same may be clarified.
 - Many units with no active power generation ($P_{gen} = 0$) are kept on in the case. These units are also providing reactive power support in the study case. The voltages in the case, therefore, may not represent the actual scenario. Such units shall therefore be kept off in the case.
- v. With above-mentioned corrections in demand and generation, the whole LGB of the study case will change. This might result in significant changes in network flows also. Therefore, it is suggested that the proposed system may be tested on realistic load-generation scenario for 2028-29, preferably the one coming out of production cost modelling studies.

NLDC Inputs on Rajasthan REZ Phase-IV and V Schemes

b) Network Loading

i. Barmer II – South Kalamb HVDC Link

- N-1 criteria is not satisfied for 400/220 kV ICTs at Barmer – II: 3000 MVA Transformation capacity for 3000 MW RE generation
- No filter arrangement modelled at HVDC terminal at Barmer - II
- On the outage of a 3000 MW Bipole of HVDC Barmer – II – South Kalamb, the loading of 765/400 kV ICTs at Barmer – I is seen to be N-1 non-compliant
- Three bus sections have been created at 400 kV Muradabad/South Kalamb station. Following issues are observed in the base case.
 - 2x1500 MVA ICTs at Section – 3 are loaded up to N-1 limit. Loading after N-1: 1490 MVA (Scenario – 4) & 1460 MVA (Scenario – 7).
- On the outage of a 1500 MW monopole of HVDC Barmer – II – South Kalamb, the loading of lines and ICTs around Barmer – II is seen to be N-1 compliant

ii. Bikaner V – Begunia HVDC Link

- N-1 compliance not satisfied for 400/220 kV ICTs at Bikaner – V: 2000 MVA Transformation capacity for 2000 MW RE generation
- STATCOM with 9999 MVA capacity connected to Bikaner – V station as filter arrangement
- On the outage of a 1500 MW monopole or 3000 MW Bipole of HVDC Bikaner – V – Begunia, the loading of lines and ICTs around Bikaner – V/Bhadla - IV is seen to be N-1 compliant
- 765/400 kV ICTs at Gopalpur are loaded up to the full rating and are N-1 non-compliant in the base case itself for the considered bulk load

iii. Highly Loaded Lines/ICTs

- Several ICTs in Rajasthan are loaded much above the N-1 limit (Deedwana - PG, Hindaun, Hirapura, Chittorgarh, Kota - PG).
- In the base case, several ICTs in WR are violating N-1 criteria (765/400 kV Jamnagar - ISTS, 765/400 kV Indore – PG, 400/220 kV Mandsaur, 400/220 kV Hazira, 400/132 kV Kirnapur – MP, 400/220 kV Navi Mumbai etc.)
- Presently, critical loading in 400 kV Farakka – Kahalgaon D/C, 400 kV New Ranchi – PPSP D/C and ICTs in West Bengal are already observed in real-time. These elements are severely loaded in the base case also (2028-29 timeframe).

Details of Lines and ICTs loaded around/more than the N-1 loading limit in NR, WR & ER are given in Annexure – 2.

NLDC Inputs on Rajasthan REZ Phase-IV and V Schemes

iv. Other Comments

- 765 kV Narela ISTS – Meerut loaded beyond 2600 MW in base case. Loading exceeds 3100 MW under the outage of 765 kV Moga – Meerut etc.
- 765 kV Aligarh – G.Noida touches 3200 MW (Scenario – 4) on N-1 of 765 kV Aligarh – Kanpur.
- RATE-2 limits of several lines are not available. The same may be corrected in the cases shared by CTUIL.

c) Voltages

- Very low voltages have been experienced at various nodes in the system including in NR RE complex and ER load centres.
- The voltages under N-1 condition (depleted network conditions) will further dip.
- No reactive power consumption considered for bulk loads considered at Paradeep & Goplapur station.
 - With 0.95 lag power factor voltages of 400 kV Gopalpur, 400 kV Paradeep – ISTS and STU stations go below 0.9 pu in steady state. 765 kV level voltage of Paradeep – ISTS & Goplapur – ISTS go below 710 kV. ***This is the case when a dynamic FACTS device of the capacity of 9999 MVA is connected at Begunia – HVDC as filter arrangement.***
 - When a 50% fixed shunt is connected at 400 kV Begunia HVDC (in place of 9999 MVA STATCOM) and with 0.95 pf operation of bulk loads at Paradeep and Gopalpur, voltage collapse is seen in the complex.***

765 kV Level

Name	Base kV	Bus No.	Voltage (p.u.)	Voltage (kV)
MANGALORE	765	528045	0.95	729
BARMER1	765	167883	0.96	738
SIROHI	765	167492	0.97	742
LALITPUR	765	177710	0.97	742
TALBEHAT	765	177715	0.97	744
TUTICORIN	765	548098	0.97	746
PARADEEP	765	428017	0.98	746
KISHENPUR	765	117706	0.98	748
GOPALPUR	765	428020	0.98	748
DUBURI_765	765	428024	0.98	748
FATEHG-4	765	167482	0.98	750

NLDC Inputs on Rajasthan REZ Phase-IV and V Schemes

400 kV Level

Name	Base kV	Bus No.	Voltage (p.u.)	Voltage (kV)
BARMER-II	400	164885	0.94	378
MANGALORE	400	524045	0.94	375
TUTICORIN	400	544098	0.95	382
JEY-BOL_FSC	400	424005	0.93	374
MUL	400	354129	0.95	381

d) Short Circuit Ratio (SCR)

Bus/Station	Connected RE/LCC HVDC Capacity	Fault Level	SCR (Without Current Contribution of immediate RE to be Connected)
220 kV Barmer - II	3000 MW RE	11906 MVA	3.96
400 kV Barmer - II	6000 MW RE + 6000 MW LCC HVDC	23606 MVA	3.93 1.97* (*Considering the requirement of system strength for both LCC HVDC and RE plant)
220 kV Bikaner - V	2000 MW RE	9470 MVA	4.7
400 kV Bikaner – V	4000 MW RE + 6000 MW LCC HVDC	23152 MVA	5.7 2.3* (*Considering the requirement of system strength for both LCC HVDC and RE plant)
220 kV Bhadla - IV	1000 MW RE	7956 MVA	7.9
400 kV Bhadla - IV	2000 MW RE	23586 MVA	11.8
400 kV Begunia HVDC	6000 MW LCC HVDC	27867 MVA	4.5

e) HVDC Schemes

02 nos. 6000 MW LCC based HVDCs have been proposed for evacuation of bulk power from Rajasthan.

±800 kV, 6000 MW HVDC Barmer-II - South Kalamb – 1000 kMs

±800 kV, 6000 MW HVDC Bikaner-V - Begunia – 1900 kMs

- i. The rationale for terminating these HVDCs in WR (Maharashtra) and ER (Odisha) may be provided.

NLDC Inputs on Rajasthan REZ Phase-IV and V Schemes

Further, as highlighted in section (a) on Load-Generation Balance, the LGB of the case is subject to change with the proposed corrections which may lead to requirement of new transmission lines in other locations than the one currently proposed.

The production cost modelling studies duly factor in the RRO targets for each states, RE integration targets, flexibility requirements, merit order etc. Therefore, it is suggested that the study scenario may be finalized based on the output of the production cost modelling studies for the same timeframe. This LGB will itself indicate the paths/corridors along with the system augmentation is required.

- ii. The overload capability and reverse power capability of both the HVDCs has not been defined and kept open. As HVDC are planned for bulk evacuation of RE power, the outage of single or both poles of the HVDC may lead to curtailment in RE generation. This curtailment may be avoided/minimized by specifying the continuous and transient **ambient temperature-based overload capability** for the proposed HVDCs.

Similar reverse power capability may also be specified for the proposed HVDCs.

- iii. With the proposed HVDCs, there will 03 nos. 6000 MW HVDC terminals in Rajasthan in close vicinity. The system strength of the Rajasthan RE complex is already on the lower side. With a large number of power electronic controllers (RE + HVDCs) in close proximity, there is a very high probability of controller interactions in the complex. Such interactions are already being observed in the Rajasthan RE complex with ~20 GW of RE capacity at present.

It is suggested that controller interaction studies (screening and if required detailed) may be carried out (using generic models for future capacities) to identify the potential interaction issues in the complex.

- iv. The feature of power oscillation damping is also important for LCC based HVDC and shall be suitably added in specifications. The provision of tuning of POD based on measurement data may also be specified.
 - v. At least one VSC based HVDC may be planned in the complex keeping in view the black start requirements. The VSC based HVDC would also provide steady state and dynamic reactive power support.
 - vi. Filter sizing becomes important with low SCR at converter stations. Keeping in view the low SCR/fault level, it is proposed that the filter sizing shall be granular to avoid wide voltage variations during switching.
- f) **Name of New Substations** – It is requested to assign unique name for envisaged RE pooling stations. As several pooling substations with identical names but different numerical suffixes have been planned in the same district along with multiple

NLDC Inputs on Rajasthan REZ Phase-IV and V Schemes

interconnections, there is a chance of miscommunication during real-time operation among multiple constituents. For clarity of operation and ensuring that names of different substations are easily distinguished from each other, it is suggested that after finalization of the exact location, the ISTS substations are uniquely named as per the geographical name of the nearest location like village or taluk.

- g) Dynamic Simulation Studies** – With more than 89,000 MW solar generation and three HVDC terminals in Rajasthan in close vicinity, it is important that stability aspects are also studied in detail at the planning stage. The generation is also getting evacuated through large EHV lines and hence, transient stability analysis also becomes important. Therefore, it is suggested that the results of the dynamic simulation studies may be shared.
- h) Study cases for other scenarios** – The study cases for evening and solar peak scenario (Feb and June) have been shared. The study cases for balance 06 scenarios especially the off-peak cases may also be shared so that voltage related issues may also be examined in detail.

NLDC Inputs on Rajasthan REZ Phase-IV and V Schemes

Annexure - 1

From Bus No	From Bus Name	From Bus Voltage	To Bus No	To Bus Name	To Bus Voltage	CKT	LOADING IN MVA	RATING IN MVA	PERCENT LOADING
199	TARAPR-3&4	21	374017	TAPS4	400.00*	1	417.3	635	65.7
199	TARAPR-3&4	21	374017	TAPS4	400.00*	2	417.4	635	65.7
12944	KANDAL	400.00*	26456	CHIMANGAON13	132	1	79.9	111	72
112235	KISHENPUR	220.00*	114422	KISHENPUR	400	1	331.4	315	105.2
112235	KISHENPUR	220.00*	114422	KISHENPUR	400	2	331.4	315	105.2
112235	KISHENPUR	220.00*	114422	KISHENPUR	400	3	331.4	315	105.2
162093	DEEDWANA-42	220	164415	DEEDWANA	400.00*	1	235.5	315	74.8
162093	DEEDWANA-42	220	164415	DEEDWANA	400.00*	2	235.5	315	74.8
162207	HINDAU-4	220	164409	HINDAU-4	400.00*	1	326	315	103.5
162207	HINDAU-4	220	164409	HINDAU-4	400.00*	2	326	315	103.5
162211	HIRAPURA	220	164406	HERAPU-4	400.00*	1	288.9	315	91.7
162211	HIRAPURA	220	164406	HERAPU-4	400.00*	2	229.3	250	91.7
162211	HIRAPURA	220	164406	HERAPU-4	400.00*	3	229.3	250	91.7
162211	HIRAPURA	220	164406	HERAPU-4	400.00*	4	229.3	250	91.7
162228	CHITTOR-42	220	164428	CHITTOR4	400.00*	1	292.1	315	92.7
162228	CHITTOR-42	220	164428	CHITTOR4	400.00*	2	292.1	315	92.7
162228	CHITTOR-42	220	164428	CHITTOR4	400.00*	3	292.1	315	92.7
162271	CHABRA-2	220	164411	CHABRA-4	400.00*	1	253.2	315	80.4
162283	BASSI	220	164431	BASSI	400.00*	1	254.3	315	80.7
162283	BASSI	220	164431	BASSI	400.00*	2	254.3	315	80.7
162283	BASSI	220	164431	BASSI	400.00*	3	403.6	500	80.7
162300	KALISIND	220	164412	KALISI-4	400.00*	2	566.8	500	113.4
162329	AJMER42	220	164416	AJMER	400.00*	1	243.6	315	77.3
162329	AJMER42	220	164416	AJMER	400.00*	2	243.6	315	77.3
162914	BHINM-PG	220	164405	BHINMAL	400.00*	1	806.4	945	85.3
162919	KOTA	220.00*	164420	KOTA	400	1	310.3	315	98.5
162919	KOTA	220.00*	164420	KOTA	400	2	310.3	315	98.5
164404	BHADLA	400.00*	164456	BIKANE-4	400	1	1146.4	1714	66.9
164404	BHADLA	400.00*	164456	BIKANE-4	400	2	1146.4	1714	66.9
164419	RAPS_C4	400.00*	364021	SHUJALPR-4	400	1	665.7	857	77.7
164419	RAPS_C4	400.00*	364021	SHUJALPR-4	400	2	665.7	857	77.7
164429	CHIT-NEW	400	164492	SIROHI	400.00*	1	656.6	850	77.2
164429	CHIT-NEW	400	164492	SIROHI	400.00*	2	656.6	850	77.2
174445	MUZAFRN4	400	174905	MEERUT	400.00*	1	580.6	857	67.8
192221	KASHIPU2	220	194467	KASHIPU4	400.00*	1	226.8	315	72
192221	KASHIPU2	220	194467	KASHIPU4	400.00*	2	226.8	315	72
192221	KASHIPU2	220	194467	KASHIPU4	400.00*	3	226.8	315	72
192225	RISHIKE2	220	194426	RISHIKE4_PT	400.00*	1	208	315	66
192225	RISHIKE2	220	194426	RISHIKE4_PT	400.00*	2	158.4	240	66
352022	HAZIRA2	220	354022	HAZIRA4	400.00*	1	347.3	500	69.5
352022	HAZIRA2	220	354022	HAZIRA4	400.00*	2	347.3	500	69.5
352022	HAZIRA2	220	354022	HAZIRA4	400.00*	3	347.2	500	69.4

NLDC Inputs on Rajasthan REZ Phase-IV and V Schemes

From Bus No	From Bus Name	From Bus Voltage	To Bus No	To Bus Name	To Bus Voltage	CKT	LOADING IN MVA	RATING IN MVA	PERCENT LOADING
352140	HALWAD NEW	220	354036	HALVAD NEW	400.00*	1	230	315	73
352140	HALWAD NEW	220	354036	HALVAD NEW	400.00*	2	230	315	73
354022	HAZIRA4	400.00*	354290	SOUTH OLPAD	400	1	964.7	857	112.6
354079	JAMNAGAR	400	358079	JAMNAGAR7	765.00*	2	987.7	1500	65.8
354079	JAMNAGAR	400	358079	JAMNAGAR7	765.00*	3	987.7	1500	65.8
361336	KIRNAPUR-421	132	364049	KIRNAPUR	400.00*	1	70.4	100	70.4
361336	KIRNAPUR-421	132	364049	KIRNAPUR	400.00*	2	70.4	100	70.4
361336	KIRNAPUR-421	132	364049	KIRNAPUR	400.00*	3	70.4	100	70.4
362003	INDORE-42	220	364001	INDORE-4	400.00*	1	262.3	315	83.3
362003	INDORE-42	220	364001	INDORE-4	400.00*	2	262.3	315	83.3
362003	INDORE-42	220	364001	INDORE-4	400.00*	3	262.3	315	83.3
362003	INDORE-42	220	364001	INDORE-4	400.00*	4	262.3	315	83.3
362113	MANDSOUR-42	220	364039	MANDSAUR-4	400.00*	1	262.1	315	83.2
362113	MANDSOUR-42	220	364039	MANDSAUR-4	400.00*	2	262.1	315	83.2
362189	DATIYA NEW2	220.00*	364189	DATIYA NEW4	400	1	440.3	500	88.1
362189	DATIYA NEW2	220.00*	364189	DATIYA NEW4	400	2	440.3	500	88.1
364023	INDORE-74	400.00*	368023	INDORE-7	765	2	1506	1500	100.4
372365	NAVI-MUM220	220	374217	NAVI-MUM	400.00*	1	267.3	315	84.9
372365	NAVI-MUM220	220	374217	NAVI-MUM	400.00*	2	267.3	315	84.9
372460	VIKROLI220	220	374068	VIKROLI400	400.00*	1	330	500	66
372460	VIKROLI220	220	374068	VIKROLI400	400.00*	2	330	500	66
372460	VIKROLI220	220	374068	VIKROLI400	400.00*	3	330	500	66
374008	DHULE4	400.00*	374011	BABLESWAR	400	2	668.7	857	78
414010	KAHALGAON-B	400.00*	444019	FARAKKA	400	1	943.9	852	110.8
414010	KAHALGAON-B	400.00*	444019	FARAKKA	400	2	943.9	852	110.8
442012	KTPS220	220	444012	KOLAGHAT	400.00*	1	281	315	89.2
442012	KTPS220	220	444012	KOLAGHAT	400.00*	2	281	315	89.2
442015	CHANDITALA_N	220	444015	CHANDITALA_N	400.00*	1	283.5	315	90
442015	CHANDITALA_N	220	444015	CHANDITALA_N	400.00*	2	283.5	315	90
442015	CHANDITALA_N	220	444015	CHANDITALA_N	400.00*	3	283.5	315	90
442685	JEERAT	220	444008	JEERAT	400.00*	1	242.9	315	77.1
442685	JEERAT	220	444008	JEERAT	400.00*	2	242.9	315	77.1
442685	JEERAT	220	444008	JEERAT	400.00*	3	242.9	315	77.1
442685	JEERAT	220	444008	JEERAT	400.00*	4	242.9	315	77.1
444074	PPSP_NEW	400.00*	444075	PURULIAPS	400	1	657.5	850	77.3
444074	PPSP_NEW	400.00*	444075	PURULIAPS	400	2	657.5	850	77.3
444074	PPSP_NEW	400	474047	RANCHI-NEW	400.00*	1	881.1	1093	80.6
444074	PPSP_NEW	400	474047	RANCHI-NEW	400.00*	2	881.1	1093	80.6
452001	DURGAPUR TPS	220	454001	DURGAPUR TPS	400.00*	1	379.2	500	75.8
452001	DURGAPUR TPS	220	454001	DURGAPUR TPS	400.00*	2	379.2	500	75.8
452002	MEJIA-B	220	454002	MEJIA-B	400.00*	1	237.4	315	75.4
452002	MEJIA-B	220	454002	MEJIA-B	400.00*	2	237.4	315	75.4

NLDC Inputs on Rajasthan REZ Phase-IV and V Schemes

Annexure – 2

From Bus No	From Bus Name	From Bus Voltage	To Bus No	To Bus Name	To Bus Voltage	CKT	LOADING IN MW	RATING IN MVA	PERCENT LOADING
199	TARAPR-3&4	21	374017	TAPS4	400.00*	1	417.3	635	65.7
199	TARAPR-3&4	21	374017	TAPS4	400.00*	2	417.4	635	65.7
12944	KANDAL	400.00*	26456	CHIMANGAON13	132	1	79.9	111	72
112235	KISHENPUR	220.00*	114422	KISHENPUR	400	1	331.4	315	105.2
112235	KISHENPUR	220.00*	114422	KISHENPUR	400	2	331.4	315	105.2
112235	KISHENPUR	220.00*	114422	KISHENPUR	400	3	331.4	315	105.2
162093	DEEDWANA-42	220	164415	DEEDWANA	400.00*	1	235.5	315	74.8
162093	DEEDWANA-42	220	164415	DEEDWANA	400.00*	2	235.5	315	74.8
162207	HINDAU-4	220	164409	HINDAU-4	400.00*	1	326	315	103.5
162207	HINDAU-4	220	164409	HINDAU-4	400.00*	2	326	315	103.5
162211	HIRAPURA	220	164406	HERAPU-4	400.00*	1	288.9	315	91.7
162211	HIRAPURA	220	164406	HERAPU-4	400.00*	2	229.3	250	91.7
162211	HIRAPURA	220	164406	HERAPU-4	400.00*	3	229.3	250	91.7
162211	HIRAPURA	220	164406	HERAPU-4	400.00*	4	229.3	250	91.7
162228	CHITTOR-42	220	164428	CHITTOR4	400.00*	1	292.1	315	92.7
162228	CHITTOR-42	220	164428	CHITTOR4	400.00*	2	292.1	315	92.7
162228	CHITTOR-42	220	164428	CHITTOR4	400.00*	3	292.1	315	92.7
162271	CHABRA-2	220	164411	CHABRA-4	400.00*	1	253.2	315	80.4
162283	BASSI	220	164431	BASSI	400.00*	1	254.3	315	80.7
162283	BASSI	220	164431	BASSI	400.00*	2	254.3	315	80.7
162283	BASSI	220	164431	BASSI	400.00*	3	403.6	500	80.7
162300	KALISIND	220	164412	KALISI-4	400.00*	2	566.8	500	113.4
162329	AJMER42	220	164416	AJMER	400.00*	1	243.6	315	77.3
162329	AJMER42	220	164416	AJMER	400.00*	2	243.6	315	77.3
162914	BHINM-PG	220	164405	BHINMAL	400.00*	1	806.4	945	85.3
162919	KOTA	220.00*	164420	KOTA	400	1	310.3	315	98.5
162919	KOTA	220.00*	164420	KOTA	400	2	310.3	315	98.5
164404	BHADLA	400.00*	164456	BIKANE-4	400	1	1146.4	1714	66.9
164404	BHADLA	400.00*	164456	BIKANE-4	400	2	1146.4	1714	66.9
164419	RAPS_C4	400.00*	364021	SHUJALPR-4	400	1	665.7	857	77.7
164419	RAPS_C4	400.00*	364021	SHUJALPR-4	400	2	665.7	857	77.7
164429	CHIT-NEW	400	164492	SIROHI	400.00*	1	656.6	850	77.2
164429	CHIT-NEW	400	164492	SIROHI	400.00*	2	656.6	850	77.2
174445	MUZAFRN4	400	174905	MEERUT	400.00*	1	580.6	857	67.8
192221	KASHIPU2	220	194467	KASHIPU4	400.00*	1	226.8	315	72
192221	KASHIPU2	220	194467	KASHIPU4	400.00*	2	226.8	315	72
192221	KASHIPU2	220	194467	KASHIPU4	400.00*	3	226.8	315	72
192225	RISHIKE2	220	194426	RISHIKE4_PT	400.00*	1	208	315	66
192225	RISHIKE2	220	194426	RISHIKE4_PT	400.00*	2	158.4	240	66
352022	HAZIRA2	220	354022	HAZIRA4	400.00*	1	347.3	500	69.5
352022	HAZIRA2	220	354022	HAZIRA4	400.00*	2	347.3	500	69.5

NLDC Inputs on Rajasthan REZ Phase-IV and V Schemes

From Bus No	From Bus Name	From Bus Voltage	To Bus No	To Bus Name	To Bus Voltage	CKT	LOADING IN MW	RATING IN MVA	PERCENT LOADING
352022	HAZIRA2	220	354022	HAZIRA4	400.00*	3	347.2	500	69.4
352140	HALWAD NEW	220	354036	HALVAD NEW	400.00*	1	230	315	73
352140	HALWAD NEW	220	354036	HALVAD NEW	400.00*	2	230	315	73
354022	HAZIRA4	400.00*	354290	SOUTH OLPAD	400	1	964.7	857	112.6
354079	JAMNAGAR	400	358079	JAMNAGAR7	765.00*	2	987.7	1500	65.8
354079	JAMNAGAR	400	358079	JAMNAGAR7	765.00*	3	987.7	1500	65.8
361336	KIRNAPUR-421	132	364049	KIRNAPUR	400.00*	1	70.4	100	70.4
361336	KIRNAPUR-421	132	364049	KIRNAPUR	400.00*	2	70.4	100	70.4
361336	KIRNAPUR-421	132	364049	KIRNAPUR	400.00*	3	70.4	100	70.4
362003	INDORE-42	220	364001	INDORE-4	400.00*	1	262.3	315	83.3
362003	INDORE-42	220	364001	INDORE-4	400.00*	2	262.3	315	83.3
362003	INDORE-42	220	364001	INDORE-4	400.00*	3	262.3	315	83.3
362003	INDORE-42	220	364001	INDORE-4	400.00*	4	262.3	315	83.3
362113	MANDSOUR-42	220	364039	MANDSAUR-4	400.00*	1	262.1	315	83.2
362113	MANDSOUR-42	220	364039	MANDSAUR-4	400.00*	2	262.1	315	83.2
362189	DATIYA NEW2	220.00*	364189	DATIYA NEW4	400	1	440.3	500	88.1
362189	DATIYA NEW2	220.00*	364189	DATIYA NEW4	400	2	440.3	500	88.1
364023	INDORE-74	400.00*	368023	INDORE-7	765	2	1506	1500	100.4
372365	NAVI-MUM220	220	374217	NAVI-MUM	400.00*	1	267.3	315	84.9
372365	NAVI-MUM220	220	374217	NAVI-MUM	400.00*	2	267.3	315	84.9
372460	VIKROLI220	220	374068	VIKROLI400	400.00*	1	330	500	66
372460	VIKROLI220	220	374068	VIKROLI400	400.00*	2	330	500	66
372460	VIKROLI220	220	374068	VIKROLI400	400.00*	3	330	500	66
374008	DHULE4	400.00*	374011	BABLESWAR	400	2	668.7	857	78
414010	KAHALGAON-B	400.00*	444019	FARAKKA	400	1	943.9	852	110.8
414010	KAHALGAON-B	400.00*	444019	FARAKKA	400	2	943.9	852	110.8
442012	KTPS220	220	444012	KOLAGHAT	400.00*	1	281	315	89.2
442012	KTPS220	220	444012	KOLAGHAT	400.00*	2	281	315	89.2
442015	CHANDITALA_N	220	444015	CHANDITALA_N	400.00*	1	283.5	315	90
442015	CHANDITALA_N	220	444015	CHANDITALA_N	400.00*	2	283.5	315	90
442015	CHANDITALA_N	220	444015	CHANDITALA_N	400.00*	3	283.5	315	90
442685	JEERAT	220	444008	JEERAT	400.00*	1	242.9	315	77.1
442685	JEERAT	220	444008	JEERAT	400.00*	2	242.9	315	77.1
442685	JEERAT	220	444008	JEERAT	400.00*	3	242.9	315	77.1
442685	JEERAT	220	444008	JEERAT	400.00*	4	242.9	315	77.1
444074	PPSP_NEW	400.00*	444075	PURULIAPS	400	1	657.5	850	77.3
444074	PPSP_NEW	400.00*	444075	PURULIAPS	400	2	657.5	850	77.3
444074	PPSP_NEW	400	474047	RANCHI-NEW	400.00*	1	881.1	1093	80.6
444074	PPSP_NEW	400	474047	RANCHI-NEW	400.00*	2	881.1	1093	80.6
452001	DURGAPUR TPS	220	454001	DURGAPUR TPS	400.00*	1	379.2	500	75.8
452001	DURGAPUR TPS	220	454001	DURGAPUR TPS	400.00*	2	379.2	500	75.8
452002	MEJIA-B	220	454002	MEJIA-B	400.00*	1	237.4	315	75.4
452002	MEJIA-B	220	454002	MEJIA-B	400.00*	2	237.4	315	75.4

NRLDC Inputs on Joint study meeting for Transmission system for evacuation of RE power from Rajasthan (REZ Ph-V (Part-1 : 4 GW), REZ Ph-IV (Part-5 : 6GW) & REZ Ph-IV (Part-6 : 6GW))
SCENARIO 2028-2029

1. SCR of following Station are below 5 considering Xsource of IBRs as 999pu :

Bus	Name	Generation connected	Fault MVA	SCR
163482	[FATEHG4-SPL	3480	12172.67	3.50
162885	[BARMER-II HV	3000	11803.85	3.93
162883	[BARMER-I	4000	16338.1	4.08
162495	[BIKANER-2	4460	18780.77	4.21
162399	[ESSEL	750	3174.92	4.23
162492	[SIROHI	2000	9235.59	4.62
162002	[BIKANER-V	2000	9805.4	4.90

- HVDC Barmer-II – South Kalamb(1000km), HVDC Bikaner-V – Begunia (1900km) are proposed as a LCC. Low SCR at Barmer-II and Bikaner-V may not be suitable for LCC HVDC. Either additional connectivity/Synchronous Condensor may be thought of in above station to improve SCR (minimum 5) or VSC HVDC in place of LCC HVDC may be considered.

2. Buses with high angular difference >20 degree under N-0/N-1

From	To	Ang. Diff (N-0)	Ang. Diff (N-1)
164774 Khetri	164495 Bikaner-2	23	25
164404 Bhadla	164456 Bikaner-4	19	24.2
167497 Sikar New	167484 Bhadla-3	16.8	21.2
167458 Bikaner-NW	137703 Moga	20.2	25
164404 Bhadla	164400 Merta	22.2	25.5
167080 NeemR(NEW)	167505 Bikaner-3	18.3	22.2

- As per CEA TPC 3.14.12, Stability studies may be carried out for the above lines.

3. Many thermal plants are observed to be **switched off during Solar time** and **Switched on (close to rated IC) during evening hours**. The philosophy may be reviewed/confirmed else sufficient BESS/Energy storage capacity may be thought off to absorb the excess RE generation during day time. Some plants in NR are listed below:

Plant/Unit (MW)	Solar Peak(MW)	Evening peak(MW)
Kalisindh (2 X 600)	0	1020
Chabra SC Unit# 6 (660)	0	560
Kanpur Unit #1 (600)	0	560
Lalitpur (3 X 660)	0	1120
Ghatampur Unit#3 (660)	0	560
Obra Unit#3,4,5 (600)	0	540
Unchahar Unit#4,5 (420)	0	170
Ropar Unit#2,3,4 (630)	0	540

4. Approx 4000 MW Solar generation in Rajasthan State control area is not being dispatched in solar peak case. Rajasthan solar dispatched is ~8000 WM against ~12000MW Installed intrastate Solar capacity in Basecase.
5. No of ICTs at 164505 Bikaner-3 may be checked: 1 X 6000MVA + 5 X 1500 MVA

NRLDC Inputs on Joint study meeting for Transmission system for evacuation of RE power from Rajasthan (REZ Ph-V (Part-1 : 4 GW), REZ Ph-IV (Part-5 : 6GW) & REZ Ph-IV (Part-6 : 6GW))

6. Giral plant is observed to be running. May be rectified.
7. At 164001 Bikaner-V, STATCOM is considered in service with Shunt Max capacity of 9999 MVA and Bridge Max capacity of 9999 MW. After taking out the STATCOM, voltage is dipping from 400 kV to 378kV.

Grid-India (NLDC/NRLDC) observations on Joint study meeting held on 09.05.24 to deliberate & finalize the Transmission system for evacuation of power from Rajasthan REZ Ph-V (Part-1 :4 GW) [Sirohi/Nagaur] Complex

1. Demand considered at all India level as well as in some of the regions in Feb 2027 period is even more than the peak demand of the year 2026-27 as per 20th EPS report. The actual average demand met during Feb'24 solar hours has also been provided for reference. It appears that the growth demand considered in NR, WR, ER, and at all India level is on the higher side. Same may be reviewed.

CTU reply :

S.No	Region	Demand considered in PSSE files (GW)	Peak demand in Feb'24 (in Winter Solar peak hours) (GW)	Anticipated Peak demand in Feb'27 (in Winter Solar peak hours) (@7% CAGR) (GW)	Demand considered in Revised PSSE files (in Winter Solar peak hours) (GW)
1	NR	78.2	63.8	78.2	79.5
2	WR	98.8 (Considering additional load of GH/BKC-8.2GW)	73.3	89.8	99.8 (Considering additional load of GH/BKC-8.2GW)
3	ER	29.5 (considering additional load of GH-3GW, BKC:0.3GW)	22.1	27.1	29.6 (considering additional load of GH/BKC-3.3GW)
	All India	288.5	220.7	270.3	291.7 (considering additional load of GH/BKC -13GW)

* GH – Green Hydrogen, BKC – Bulk Consumer

From the historical data (Feb'23 to Feb'24) analysis, load growth of about 7 % is observed in winter solar peak hours in All India & most of the regions. Based on extrapolation of above load growth, expected load of Feb'27 (in winter solar peak hours) is indicated on the table above which is considered in PSS/E file. Further additional demand from Green Hydrogen, Bulk consumer is also considered the studies (included in total demand figure).

- Green hydrogen / Bulk loads considered in ER are as under
 - 1500MW: Paradeep,
 - 1500MW: Gopalpur,
 - 180MW: Vedanta & 100MW: IPCL
- Major Bulk loads/ Green hydrogen considered in WR are as under
 - 4500MW: MUL Gujarat (including 1500MW Green Hydrogen)
 - 3700MW: Jamnagar/Jam Khambaliya Complex

2. Thermal generation has been uniformly backed down to 55% (especially in NR and WR) at most of the stations. It is suggested that the generation may be scaled as per the merit order. Cheaper thermal generating stations like Singrauli and Rihand are also running at 55% level. some of the thermal generating stations (like Vadinar, Khaparkheda, Tiroda-APL etc.) are running below 40% tech min level also. Same may be clarified.

CTU reply – In revised files, merit order dispatch is being followed up to an extent and incorporated in revised PSS/E files. Dispatch of Cheaper thermal generating stations in Singrauli and Rihand complex kept higher to match LGB. .

Dispatches of major thermal generating stations of states such as Vadinar, Khaparkheda, Tiroda-APL, Chandrapur have been changed to 40%. Further, generation with very low dispatches has also been switched off.

3. Many units with no active power generation ($P_{gen} = 0$) are kept on in the case. These units are also providing reactive power support in the study case. The voltages in the case, therefore, may not represent the actual scenario. Such units shall therefore be kept off in the case

CTU reply - Incorporated (all such units are now kept off)

4. With above-mentioned corrections in demand and generation, the whole LGB of the study case will change. This might result in significant changes in network flows also. Therefore, it is suggested that the proposed system may be tested on realistic load generation scenario for 2026-27, preferably the one coming out of production cost modelling studies.

CTU reply – Modified PSSE files are enclosed. It is observed that with above modifications in LGB, there will be no impact on the proposed transmission scheme for evacuation of power from Rajasthan REZ Ph-V (Part-1 :4 GW) [Sirohi/Nagaur] Complex.

5. There is significant amount of power flow towards eastern region during solar hours resulting in very high loading and low voltages in the ER grid. It is suggested that system augmentation in ER may be taken up on priority so that the required system gets commissioned with the associated RE generation (2026-27 timeframe).

CTU reply - Reconductoring of 400kV Farakka – Kahalgaon & Talcher – Meramundali 400kV D/c lines in under advance stage of stakeholder consultation. Other system, if required in ISTS would be taken up on priority

6. Several ICTs in Rajasthan are loaded much above the N-1 limit (Deedwana - PG, Hindaun, Hirapura, Chittorgarh, Kota – PG etc.

CTU reply – Most of the ICTs i.e. Deedwana, Hindaun, Heerapura, Chittorgarh are getting overloaded in planning studies due to the increase in load of Rajasthan. Some of the ICTs are even critically loaded in present time frame and issue of high loading is deliberated in various OCC/NRPC meetings. In above meetings, RVPN is already requested to carry out 400/220kV ICT augmentation works or feeder reconfiguration to relive ICT loadings.

Loading of 400/220kV Kota (PG) ICT is higher due to low dispatch of KTPS generation units. Loading of Bhinmal ICT is higher in Sc-7 due to increasing demand.

Above ICT loading will be reviewed in consultation with NRLDC w.r.t real time loading and Grid-India operational feedback report and necessary augmentation will be carried out in ISTS, if required, as per the present practice.

7. In the base case, several ICTs in WR are violating N-1 criteria (765/400 kV Jamnagar - ISTS, 765/400 kV Indore – PG, 400/220 kV Mandsaur, 400/220 kV Hazira, 400/132 kV Kirnapur – MP, 400/220 kV Navi Mumbai etc.)

CTU reply – In Jamnagar Complex, applications from bulk consumer to the tune of about 1800MW has been received, whereas in PSSE file load of 3700MW has been considered. Accordingly, ICT augmentation would be taken up with receipt of further applications.

Augmentation of Transformation capacity at 765/400kV Indore S/s by 1x1500MVA ICT (3rd) [terminated on 400kV Bus section A with Indore & Khandwa 400kV D/c lines] has been agreed in CMETS-WR & is presently under implementation. On other section, under contingency of any line, sectionaliser can be closed to reduce loading on 765/400kV ICT. To resolve issue of overloading at Mandsaur, 1x500MVA, 400/220kV ICT (3rd) at Mandsaur S/s has been planned & is being implemented by MPPTCL. Establishment of 400/220kV Hazira-II (GIS) S/s through Installation of 1x500MVA, 400/220kV ICT has been agreed for drawl of power at Hazira in WR-CMETS meeting to reduce overloading on Hazira S/s. ICT augmentation or shifting of proposed Data Centres load at Navi Mumbai is being taken up with MSETCL.

8. Critical loading in 400 kV Farakka – Kahalgaon D/C, 400 kV New Ranchi – PPSP D/C and ICTs in West Bengal is already observed in real-time during solar hours. These elements are severely loaded in the base case also (2027 timeframe)

CTU reply – Reconductoring of 400kV Farakka – Kahalgaon D/c line in under active stages of discussion.

New generation are expected in Purulia area viz. Raghunathpur (2x660MW) and DTPS-Waria (1x800MW). With this, the loading on Ranchi – New Purulia is expected to be relieved. Nevertheless, analysis of loading on Ranchi – New Purulia 400kV D/c line would be taken up for discussion in future.

9. 765 kV Ghiror – Aligarh is completely off-loaded (5 MW loading in each ckt)

CTU reply - 765 kV Ghiror – Aligarh D/c line is formed through LILO of existing 765V Aligarh-Orai D/c line at Ghiror S/s for optimal utilization of above line in solar maximized scenario. It is to be mention that 765kV Ghiror and Aligarh both are the major substations for dispersal of RE power towards load centers in UP. In view of huge RE injection each at 765kV Aligarh & 765kV Ghiror S/s, 765 kV Ghiror – Aligarh line gets off-loaded. However in other than solar maximized scenario, line loading is more than loading in solar maximized scenario In future studies and schemes, efforts to be made for optimal utilization of Aligarh-Ghiror section.

10. 765 kV Aligarh – G.Noida touches 2800 MW (Scenario – 7) & 3400 MW (Scenario – 4) on N-1 of 765 kV Aligarh – Kanpur

11. **CTU reply** – In the planning studies, maximum loading of 3500MW is considered on 765kV D/c lines in N-1 contingency in case other parameters i.e. angular separation, bus voltages are within stipulated limit as per transmission planning criteria. In 765kV Aligarh – G.Noida line, angular separation is within limit and voltages are in order on N-1 contingency of 765 kV Aligarh – Kanpur
12. Angular difference between 765 kV Bikaner and 765 kV Moga touches 26° under N-1 (NLDC). Additionally, some Buses i.e. 400kV Bikaner-II – Khetri, 765kV Bhadla-III-Sikar-II, 765kV Bikaner-III-Neemrana-2 have angular difference >20 degree under N-0/N-1 (NRLDC)

CTU reply – The loading of 765 kV Bikaner and 765 kV Moga will gradually increase in next 3 years in planning studies due to increasing RE injection at Bikaner. Stability studies shall be carried out for 765kV Bikaner-Moga D/c line and 400kV Bikaner-II – Khetri D/c line for 2027 timeframe, once dynamics file is prepared. Presently there are convergence issues being faced in dynamic simulation for 2027 scenario for which consultation of IIT Mumbai is also under progress.

Angular separation of 765kV Bhadla-III-Sikar-II & 765kV Bikaner-III-Neemrana-2 is below 20 degree in N-1 contingency in 2027 timeframe files (angular separation >20 degree in N-1 contingency in 2029 timeframe files for which Stability studies shall be carried out once dynamics file is prepared)

13. RATE-2 limits of several lines are not available. The same may be corrected in the cases shared by CTUIL.

CTU reply - Incorporated.

14. Very low voltages have been experienced at various nodes in the system including in NR RE Complex. ii. The voltages under N-1 condition (depleted network conditions) will further dip.

CTU reply – In revised PSSE files most of the bus voltage are within limits of 0.97-1.03 pu on all 400kV and 765kV buses in NR considering switching on/off of bus and line reactors (switchable) wherever required.

15. Low SCR of Sirohi PS (4.2)

CTU reply – Short circuit Ratio of 400kV and 220kV of Sirohi PS is about 8.7 and 4.2 respectively. No additional connectivity will be granted beyond 2 GW at 220kV level of Sirohi PS with proposed system (400/220kV ICTs at Sirohi PS+765kV Sirohi-Mandsaur). With growing interconnection and planned network in future, SCR at Sirohi may increase.

16. Dynamic Simulation Studies – With more than 62,000 MW solar generation and three HVDC terminals in Rajasthan in close vicinity, it is important that stability aspects are also studied in detail at the planning stage. The generation is also getting evacuated through large EHV lines and hence, transient stability analysis also becomes important. Therefore, it is suggested that the results of the dynamic simulation studies may be shared.

CTU reply – Comment is pertaining to 2029 timeframe files, however dynamics file is under preparation. Presently there are convergence issues being faced in dynamic simulation for 2027 scenario for which consultation of IIT Mumbai is also under progress

17. Study cases for other scenarios – The study cases for evening and solar peak scenario (Feb and June) have been shared. The study cases for balance 06 scenarios especially the off-peak cases may also be shared so that voltage related issues may also be examined in detail.

CTU reply – As per the comments, Scenario 8 (evening peak) & Scenario 9 (night off peak) along with revised solar maximized scenario (SC-4& 7) files is attached.

18. Giral plant is observed to be running. May be rectified (NRLDC).

CTU reply – Giral plant is now kept off

NLDC Inputs on Revised Scheme for Rajasthan REZ Phase- V RE

1. Schemes proposed for:

a) Rajasthan REZ Ph-V (Part-1: 4GW) (Sirohi/Nagaur Complex)

S.No	Transmission Scheme	RE Potential	Status
A	Under Bidding/ Approved		
1	Rajasthan REZ Ph-IV (Part-1 :7.7GW) (Bikaner Complex)	14 GW (Solar 14GW, BESS:6GW) Bikaner-II : 3.7GW Bikaner-III: 4GW	Awarded
2	Rajasthan REZ Ph-IV (Part-2 :5.5GW) (Jaisalmer/Barmer Complex)	5.5GW (Solar) Fatehgarh-IV: 4 GW Barmer-I: 1.5 GW	Under Bidding
3	Rajasthan REZ Ph-IV (Part-3 :6GW) (Bikaner Complex)	6 GW (Solar) Bikaner-IV:6GW	Under Bidding (Recently approved in NCT)
B	Planned/Under Planning		
1	Rajasthan REZ Ph-IV (Part-4 :3.5GW) (Jaisalmer/Barmer Complex)	3.5 GW (Solar) Fatehgarh-IV: 1 GW Barmer-I: 2.5 GW	Recently approved in NCT
2	Rajasthan REZ Ph-IV (Part-5 : 6GW) (Barmer Complex)	6 GW (Solar) Barmer-II : 6GW	Timeframe : 2029 (HVDC)
3	Rajasthan REZ Ph-IV (Part-6 : 6GW) (Bhadla/Bikaner Complex)	6 GW (Solar) Bhadla-IV: 2 GW Bikaner-V: 4 GW*	
4	Rajasthan REZ Ph-V (Part-1: 4GW) (Sirohi/Nagaur Complex)	4 GW (Solar) Sirohi: 2 GW Nagaur: 2 GW	

NLDC Inputs on Revised Scheme for Rajasthan REZ Phase- V RE

2. Inputs on AC Scheme proposed for Rajasthan REZ Ph-V (Part-1: 4GW) (Sirohi/Nagaur Complex)

Proposed System (Timeframe – 2027):

- 5x500MVA, 400/220kV ICTs at Sirohi S/s along with transformer bays
- Sirohi – Mandsaur PS 765KV D/c line (~ 320 kms) along with 330 MVAR switchable line reactor for each circuit at each end of Sirohi – Mandsaur PS 765KV D/c line
- Mandsaur PS – Khandwa (New) 765kV D/c line (~230km.) along with 240MVAR switchable line reactor for each circuit at each end of Mandsaur PS – Khandwa (New) 765 kV D/c line

a) Load-Generation Balance and IR Flows (Feb 2027 scenario)

S. No.	Region	Load (MW)	Gen. (MW)	IR Exchange (MW)	20 th EPS Peak Load in 2026-27 (MW)	Average Demand Met in Solar Hours in Feb 2024 (MW)	Peak Demand met in Feb 2024 (MW)	Peak Demand met in Feb 2023 (MW)
1	NR	79604	109895	30291	97898	59664	63791	59366
2	WR	99798	99386	-412	89547	72184	74167	76524
3	SR	79758	94860	15102	80864	62000	65591	60906
4	ER	29644	14828	-14816	37265	20000	23531	22325
5	NER	2974	2654	-320	4855	2000	2822	2849
6	All India	291780	321625	x	277201	216185	221989	211325

S. No.	State	Load (MW)	Gen. (MW)	IR Exchange (MW)	20 th EPS Peak Load in 2026-27 (MW)	Average Demand Met in Solar Hours in Feb 2024 (MW)	Peak Demand met in Feb 2024 (MW)	Peak Demand met in Feb 2023 (MW)
1	Rajasthan	19593	80348	60755	21175	17000	18058	16959

- It has been mentioned that CAGR of 7% has been considered on the peak demand met by respective states in Feb 2024 at regional and all India level. Over this, green hydrogen load of 13 GW has been considered in ER (3.3 GW) and WR (8.2 GW).
- The peak and average (in solar hours) demand met during Feb'23 and Feb'24 solar hours has been provided for reference. It appears that the CAGR of 7% considered at regional and all India level is on the higher side. Same may be reviewed.

The all India demand considered in Feb 2027 is even more than the EPS peak demand for 2026-27 + 13 GW green hydrogen load.

NLDC Inputs on Revised Scheme for Rajasthan REZ Phase- V RE

iii. Solar and Thermal Generation

- ~192724 MW solar generation considered in service at all India level
 - ~72500 MW solar generation is in service in Rajasthan
 - ~27770 MW solar generation is in service in Rajasthan
 - ~19100 MW solar generation in Khavda complex has been considered in the case

 - Still some machines (bus - 424027) with no active power generation ($P_{gen} = 0$) are kept on in the case. These units are also providing reactive power support in the study case. The voltages in the case, therefore, may not represent the actual scenario. Such units shall therefore be kept off in the case.

 - In the revised case also, some of the thermal generating stations (like 412001, 362014 etc.) are running below 40% tech min level also. Same may be clarified.

 - Further, the swing bus generation (Farakka) is negative – (-) 2500 MW which would further increase the demand by around 5000 MW.

 - The thermal generation has been backed down to 40%, 45% and 55% level. The tech min level considered in the case may be mentioned. Though, it is mentioned that merit order has been considered, but cheaper thermal generating stations like Singrauli and Rihand are running at only 65-70% level. Dispatch in Yadadri etc. is around 85%.
- iv. With above-mentioned corrections in demand and generation, the LGB of the study case will change. This might result in changes in network flows also. Therefore, it is suggested that the proposed system may be tested on realistic load-generation scenario for 2026-27, preferably the one coming out of production cost modelling studies.

The details of N-1 non-compliance, high line loading, and low voltages are provided in subsequent sections.

NLDC Inputs on Revised Scheme for Rajasthan REZ Phase- V RE

b) Network Loadings

- i. Several ICTs in Rajasthan are loaded much above the N-1 limit (Deedwana - PG, Hindaun, Hirapura, Chittorgarh, Kota – PG etc.)

CTUIL Response –

Most of the ICTs i.e. Deedwana, Hindaun, Heerapura, Chittorgarh are getting overloaded in planning studies due to the increase in load of Rajasthan. Some of the ICTs are even critically loaded in present time frame and issue of high loading is deliberated in various OCC/NRPC meetings. In above meetings, RVPN is already requested to carry out 400/220kV ICT augmentation works or feeder reconfiguration to relive ICT loadings.

Loading of 400/220kV Kota (PG) ICT is higher due to low dispatch of KTPS generation units. Loading of Bhinmal ICT is higher in Sc-7 due to increasing demand.

Above ICT loading will be reviewed in consultation with NRLDC w.r.t real time loading and Grid- India operational feedback report and necessary augmentation will be carried out in ISTS, if required, as per the present practice.

Grid-India Response

RVPNL may also be invited in the meeting and necessary augmentation required at intra-state level may also be planned at this stage.

For augmentation required at ISTS level, the planning may be carried out at this stage only.

- ii. Many lines in ER and WR-ER boundary are critically loaded in solar peak case:

- 400 kV Farakka – Kahalgaon D/C,
- 400 kV New Ranchi – PPSP D/C
- 400 kV Ranchi – Sipat D/C
- 765 kV Dharamjaigarh – New Ranchi D/C
- 400 kV Rourkela – Chaibasa D/C
- 400 kV Gaya – Maithon D/C
- ICTs in West Bengal (Kolaghat, Chanditala) and DVC (Mejia-B, Durgapur)

CTUIL Response - Re-conductoring of 400 kV Farakka – Kahalgaon & Talcher – Meramundali 400 kV D/c lines in under advance stage of stakeholder consultation. Other system, if required in ISTS would be taken up on priority.

NLDC Inputs on Revised Scheme for Rajasthan REZ Phase- V RE

Grid-India Response: There is significant amount of power flow towards eastern region during solar hours resulting in very high loading in the ER grid. The issue is also being faced in the current scenario during solar hours. It is suggested that system augmentation in ER in a holistic manner may be taken up on priority so that the required system gets commissioned with the associated RE generation (2026-27 timeframe).

v. Angular Separation:

Scenario	765 kV Sirohi – Mandsaur D/C Line	765 kV Sirohi – Risabhdeo D/C	765 kV Neemuch – Khandwa D/C	765 kV Bikaner – Moga D/C	765 kV Bikaner – Khetri D/C
Base Case	14.4° (1746 MW)	9.5° (1744 MW)	7.3° (1230 MW)	18.2° (1914 MW)	18.2° (2425 MW)
Under N-1 Contingency	17.5° (2100 MW)	12.4° (2254 MW)	9.3° (1581 MW)	24.4° (2526 MW)	20° (2995 MW)
Under N-1-1 Contingency	22.3°	17.9°	13.2°	42.45°	25.1°

- vi. HVDC Balia – Bhiwadi is running from Balia to Bhiwadi in evening peak scenario while the direction is Bhiwadi to Balia in solar peak case. The daily reversal of HVDC may also be highlighted in the scheme as reluctance is faced from asset owners regarding frequent reversal in HVDC direction.

The details of other highly loaded elements have already been shared earlier.

c) Voltages

- i. It has been observed that SLRs of various 765 kV lines have been opened in the solar peak case to maintain the voltages. These SLRs have been taken back in service in the off-peak case to avoid high voltages in the system.

However, in real-time, reluctance is faced from transmission licensees regarding switching of these SLRs. Even resistance is being faced in case of frequent switching of bus reactors.

Daily operation of SLRs to control the voltage may be mentioned explicitly in the schemes as well as included in the RfP also.

NLDC Inputs on Revised Scheme for Rajasthan REZ Phase- V RE

d) Short Circuit Ratio

Bus/Station	Connected RE Capacity	Fault Level	SCR (Without Current Contribution of immediate RE to be Connected)
220 kV Sirohi	2000 MW	8413 MVA	4.2
220 kV Merta – II	1000 MW	5734 MVA	5.7
400 kV Merta - II	1000 MW	20302 MVA	20.3

This SCR is considering the optimistic scenario that RE plants will provide 1 p.u. short circuit contribution. However, the plants are providing the same in real-time. Same may be considered while computing the SCR.

e) Dynamic Simulation Studies

With more than 72,500 MW solar generation in Rajasthan in close vicinity by Feb 2027, it is important that stability aspects are also studied in detail at the planning stage. The generation is also getting evacuated through large EHV lines and hence, transient stability analysis also becomes important. Therefore, it is suggested that the results of the dynamic simulation studies may be shared.

CTU reply – Comment is pertaining to 2029 timeframe files, however dynamics file is under preparation. Presently there are convergence issues being faced in dynamic simulation for 2027 scenario for which consultation of IIT Mumbai is also under progress.

Grid-India response – At present, even after than 15 GW RE at ISTS level in Rajasthan, the dynamic simulation studies have not been carried out. Keeping in view the multiple generation loss events in Rajasthan RE complex, it is once again requested that the studies may be carried out at the time of proposing the evacuation scheme.

f) Other Inputs:

- i. In addition to merit order dispatch, inertia would also be a deciding factor for a unit to remain switch on or off. Same may also be factored in.
- ii. Wherever long lines are getting planned like in this case 765 kV Sirohi-Mandsaur (320 Km), the range of line length may be provided as it is possible that route length may increase during commissioning phase.
- iii. Intra-state network wherever connected or proposed to be connected may be specified. All such long HVDCs and AC lines will impact Rajasthan ATC/TTC.

NLDC Inputs on Revised Scheme for Rajasthan REZ Phase- V RE

- iv. **Name of New Substations** – It is requested to assign unique name for envisaged RE pooling stations. As several pooling substations with identical names but different numerical suffixes have been planned in the same district along with multiple interconnections, there is a chance of miscommunication during real-time operation among multiple constituents. For clarity of operation and ensuring that names of different substations are easily distinguished from each other, it is suggested that after finalization of the exact location, the ISTS substations are uniquely named as per the geographical name of the nearest location like village or taluk.

CTU reply on Grid-India (NLDC/NRLDC) observations (13.06.24) on PSS/E files (2027) for Transmission system for evacuation of power from Rajasthan REZ Ph-V (Part-1 :4 GW) [Sirohi/Nagaur] Complex

- A.** It has been mentioned that CAGR of 7% has been considered on the peak demand met by respective states in Feb 2024 at regional and all India level. Over this, green hydrogen load of 13 GW has been considered in ER (3.3 GW) and WR (8.2 GW).

The peak and average (in solar hours) demand met during Feb'23 and Feb'24 solar hours has been provided for reference. It appears that the CAGR of 7% considered at regional and all India level is on the higher side. Same may be reviewed. The all India demand considered in Feb 2027 is even more than the EPS peak demand for 2026-27 + 13 GW green hydrogen load.

CTU reply :

In the meeting CTU stated that Transmission planning is carried out for peak load scenario. Considering this CTU analyzed the peak demand historical data (Feb'23 to Feb'24) for solar maximized period from which it is observed that load growth is more than 7% in winter solar peak hours in All India & most of the regions. Green Hydrogen and bulk consumer load is considered over and above to EPS demand in planning studies as EPS did not factor in such emerging loads.

S.No	Region	Peak demand in Feb'24 (in Winter Solar peak hours) (GW)	Peak demand in Feb'23 (in Winter Solar peak hours) (GW)	Growth rate (%)	Anticipated Peak demand in Feb'27 (in Winter Solar peak hours) (@7% CAGR) (GW)	Demand considered in Revised PSSE files (in Winter Solar peak hours) (GW)
1	NR	63.8	59.4	7.4	78.2	79.5
2	WR	73.3	68.7	6.7	89.8	100.5 (Considering additional load of GH/BKC-8.2GW)
3	ER	22.1	21.7	2	27.1	29.6 (considering additional load of GH/BKC-3.3GW)
	All India	220.7	200.6	10	270.3	295.5 (considering additional load of GH/BKC -13GW)

In view of that demand considered in the studies solar maximized scenario (Feb'27) as above is in order.

- B.** Still some machines (bus - 424027) with no active power generation ($P_{gen} = 0$) are kept on in the case. These units are also providing reactive power support in the study case. The voltages in the case, therefore, may not represent the actual scenario. Such units shall therefore be kept off in the case.

CTU Reply: Bus no 424027 corresponding to Vendata (4x600MW) generation in ER. Out of 4 units, one unit is for dedicated supply to OPTCL through Vedanta – Lapanga / Meramundali 400kV D/c line. The other 3 units are CPP units which are for dedicated supply to their smelter load. Therefore, these three units have been considered as ON with zero dispatch from these 3 units.

However, as decided in the meeting, a bulk load of smelter plant 1500MW (non scalable) has been added at the load bus and dispatch of 3x600MW plant has been set as 3x500MW.

- C. In the revised case also, some of the thermal generating stations (like 412001, 362014 etc.) are running below 40% tech min level also. Same may be clarified.

CTU Reply : Incorporated

- D. Further, the swing bus generation (Farakka) is negative – (-) 2500 MW which would further increase the demand by around 5000 MW.

CTU Reply: Modifications incorporated (Revised study plots with above modifications is enclosed)

- E. The thermal generation has been backed down to 40%, 45% and 55% level. The tech min level considered in the case may be mentioned. Though it is mentioned that merit order has been considered, but cheaper thermal generating stations like Singrauli and Rihand are running at only 65-70% level. Dispatch in Yadadri etc. is around 85%.

CTU Reply: Merit order dispatch is being followed up to an extent and incorporated in revised PSS/E files. Dispatch of Cheaper thermal generating stations in Singrauli and Rihand complex is already kept higher as compared to other machines.

- F. With above-mentioned corrections in demand and generation, the LGB of the study case will change. This might result in changes in network flows also. Therefore, it is suggested that the proposed system may be tested on realistic load-generation scenario for 2026-27, preferably the one coming out of production cost modelling studies.

CTU Reply: With these modifications in WR/ER LGB and other modifications in ER/WR regions, there is no impact on the proposed transmission scheme for evacuation of power from Rajasthan REZ Ph-V (Part-1 :4 GW) [Sirohi/Nagaur] Complex.

In the meeting, CTU reiterated that the proposed scheme is an Inter regional corridor (765kV Sirohi-Mandsaur D/c) which would relieve the loading of various other WR-NR Inter regional corridors i.e. 400kV RAPP-Shujalpur D/c, 400kV Zerda-Kankroli D/c, 765kV Sirohi-Rishabdeo D/c highlighted in review meetings of MOP as well as in CMETS-NR meetings.

- G. Several ICTs in Rajasthan are loaded much above the N-1 limit (Deedwana - PG, Hindaun, Hirapura, Chittorgarh, Kota – PG etc.)

RVPNL may also be invited in the meeting and necessary augmentation required at intra-state level may also be planned at this stage. For augmentation required at ISTS level, the planning may be carried out at this stage only.

CTU Reply: For intra state ICT overloading in Rajasthan, RVPN stated that for Hindaun ICT work awarded in Mar'24 (24 months schedule) and for Chittorgarh, Heerapura and Deedwana ICTs proposal is in process. CTU requested RVPN to expediate the approval of augmentation of ICTs so that same is available in 2027 timeframe.

CTU stated that Loading of 400/220kV Kota (PG) ICT is higher due to low dispatch of KTPS generation units. Loading of Bhinmal ICT is higher in Sc-7 due to increasing demand. Implementation of ICTs may take less time (18 months from approval) as compared to transmission schemes for RE evacuation (30 months from approval). It was stated that necessary augmentation (Kota and Bhinmal ICT) will be carried out in ISTS in consultation with stakeholders once space availability from TSP for Bhinmal and real time loading data from NRLDC is received for implementation in 2027 timeframe or earlier. NRLDC agreed for the same

H. Many lines in ER and WR-ER boundary are critically loaded in solar peak case:

- 400 kV Farakka – Kahalgaon D/C,
- 400 kV New Ranchi – PPSP D/C
- 400 kV Ranchi – Sipat D/C
- 765 kV Dharamjaigarh – New Ranchi D/C
- 400 kV Rourkela – Chaibasa D/C
- 400 kV Gaya – Maithon D/C
- ICTs in West Bengal (Kolaghat, Chanditala) and DVC (Mejia-B, Durgapur)

There is significant amount of power flow towards eastern region during solar hours resulting in very high loading in the ER grid. The issue is also being faced in the current scenario during solar hours. It is suggested that system augmentation in ER in a holistic manner may be taken up on priority so that the required system gets commissioned with the associated RE generation (2026-27 timeframe).

CTU Reply:. In the meeting, Grid-India stated that 400 kV Farakka – Kahalgaon D/c & 400 kV New Ranchi – PPSP D/c is overloaded in real time also. In view of RE addition in NR and WR and backing down of thermal units in ER, critical loading is observed in ER and WR-ER corridors in planning studies. In view of that strengthening is being taken up on priority with progressive addition of solar generation in next 2-3 years so that schemes in ER are available for absorption of RE power in Eastern region.

In the meeting, CTU stated that Reconductoring of some of the lines i.e. 400kV Farakka – Kahalgaon and 400kV Talcher – Meramundali is recently agreed in CMETS-ER meeting with a completion schedule of 18 months from allocation.

In real time, loading of 400 kV New Ranchi – PPSP D/c is in order as line rating was considered lower, however in planning studies loadings are higher. Reconductoring of Ranchi New – New PPSP 400kV D/c line and any other augmentation in ISTS and commensurate STU network would be identified based on the joint study meeting scheduled next month with WBSETCL. Reconductoring of 400kV Farakka – Sagardighi and 400kV Rourkela – Chaibasa 400kV D/c line would be taken up for discussion in CMETS-ER meeting (Jul'24). An additional corridor from WR to ER needs to be envisaged for relieve the loading around Ranchi area considering LGB of 2027 and 2029 timeframe and change in power flow pattern due to envisaged Green hydrogen load. Augmentation of ICTs in WBSETCL, DVC & Odisha area have been already informed to respective STUs in various CMETS-ER meetings and joint meetings. Matter will be further emphasized to STUs/DVC to take up augmentation of ICTs within time.

It was also decided that due to huge power flow in solar peak hours from NR-ER and WR-ER, and thermal backdown in ER, there may be other system strengthening requirement within ER

incl. requirement discussed above. Such transmission strengthening requirements will be planned shortly and taken up for stakeholder consultation based on study results under various scenarios.

- I. HVDC Balia – Bhiwadi is running from Balia to Bhiwadi in evening peak scenario while the direction is Bhiwadi to Balia in solar peak case. The daily reversal of HVDC may also be highlighted in the scheme as reluctance is faced from asset owners regarding frequent reversal in HVDC direction.

CTU Reply: In all planning studies, Bhiwadi is running from Balia to Bhiwadi in evening peak scenario while the direction is Bhiwadi to Balia in solar peak case.

CTU stated that they have already informed POWERGRID on above issue faced by Grid-India. In the meeting, POWERGRID stated that reversal is being carried out whenever required without any reluctance, however during reversal operation, filters switching operation is being carried out each time which may create switching surges on filters and CBs despite that no reluctance is being offered, however reversal may be carried out in an optimal manner.

Grid-India (NLDC) stated that in case reluctance comes in future for reversal of Balia-Bhiwadi HVDC, Grid constraint may arise in RE power evacuation. In Planning, it is to be mentioned that reversal will be carried out on daily basis as part of scheme to sensitize the TSP in case any reluctance faced during reversal operation in future.

NRLDC stated that at present daily reversal is not required, however reversal requirement may be on daily basis in future and therefore TSP should take up the matter with OEM for reversal operation of HVDC on daily basis, in case of any constraints.

CTU stated that Power reversal on Balia – Bhiwadi HVDC line from Bhiwadi to Balia was approved as part of scheme “Transmission System Strengthening for potential solar energy zones Phase - II(8.1GW)” considering reversal of HVDC in solar peak hour for evacuation of Ph-II RE power from Rajasthan. Therefore, basic premise for scheme was for reversal of power from Bhiwadi to Balia in peak RE period which was known by the TSP.

CTU emphasized that reversal of Balia-Bhiwadi HVDC is required in solar peak hours (Bhiwadi to Balia) and it will be continued in solar peak hours throughout the life of this HVDC. In view of that, if required POWERGRID may discuss the matter with OEM for constraint in daily reversal, if any. Accordingly, POWERGRID may take up the matter in consultation with CTU and Grid-India for deliberation in OCC/NRPC forum. POWERGRID agreed for the same.

- J. It has been observed that SLRs of various 765 kV lines have been opened in the solar peak case to maintain the voltages. These SLRs have been taken back in service in the off-peak case to avoid high voltages in the system. However, in real-time, reluctance is faced from transmission licensees regarding switching of these SLRs. Even resistance is being faced in case of frequent switching of bus reactors. Daily operation of SLRs to control the voltage may be mentioned explicitly in the schemes as well as included in the RfP also.

CTU Reply: CTU vide mail 17.06.24 enquired TSPs for reasoning and justification on above issue faced by Grid-India during line reactor switching and frequent switching of bus reactors in real time.

In the 30th CMETS-NR meeting POWERGRID informed that switching of line/bus reactor is being carried out whenever required, However constraint of bus/line Reactor switching may come in case of GIS S/s. POWERGRID also stated that there may be some delay involved during line reactor switching due to large no. operations at a time due to manual operation involvement, NGR bypassing, checking the healthiness of equipment, CB and isolator operation etc. It is also seen that frequent operation of CBs for reactors switching cause gas formation due to switching surges.

Grid-India (NLDC) stated that in planning studies line reactor switching is envisaged to control the voltage, however in operation if any reluctance faced during line reactor switching, additional reactive compensation is required to control the voltages. In view of that reactor switching requirement shall be clearly stipulated in RfP document. Based on their operational experience, line reactors should be designed for 2 switching operations in a day, however for bus reactor they will check the requirement based on operational experience. NLDC also stated that in case of delay in implementation of total transmission system for RE evacuation, overall more nos. of reactor switching operations are required, however on daily basis 2 nos. of switching operations is sufficient.

CTU also stated that in planning studies, they have also considered 2 nos. of operation for bus and line reactor switching. CTU also stated that the issue was also highlighted earlier in the Joint study meeting held on 28.12.23 and 09.01.24 as part of Transmission scheme for Rajasthan REZ Ph-IV (Part-3) scheme. In the above Joint study meeting, CEA stated that line reactor switching in RE pockets is in general operational requirement and need not to be specific for one scheme and should be carried out in real time as per Grid requirement. It was also stated that whenever such line reactor switching issue is raised by TSP in future, it should come with proper reasoning & justification from TSP side.

In view of that it was concluded that a 2 nos. of line reactor switching operations is required and matter will be referred to CTU-Engg for inclusion of reactor switching as part of new RfP/TS document. Meanwhile, NLDC will analyze the operational data for bus reactor switching and inform the same to CTU.

K. Low SCR of Sirohi PS (4.2)

CTU reply – Short circuit Ratio of 400kV and 220kV of Sirohi PS is about 8.7 and 4.2 respectively. For low SCR at Sirohi PS(4.2), CTU stated that no additional connectivity will be granted beyond 2 GW at 220kV level of Sirohi PS with proposed system (400/220kV ICTs at Sirohi PS+765kV Sirohi - Mandsaur). With growing interconnection and planned network in future, SCR at Sirohi may increase.

Transmission scheme is already under planning for RE evacuation from Sirohi/Jalore/Sanchore complex and taken up for stakeholder consultation in next 1-2 months.

L. At present, even after than 15 GW RE at ISTS level in Rajasthan, the dynamic simulation studies have not been carried out. Keeping in view the multiple generation loss events in Rajasthan RE complex, it is once again requested that the studies may be carried out at the time of proposing the evacuation scheme.

- M. Angular difference between 765 kV Bikaner and 765 kV Moga touches 26° under N-1 and 42.4 degree under N-1-1.
- N. In addition to merit order dispatch, inertia would also be a deciding factor for a unit to remain switch on or off. Same may also be factored in.

CTU Reply: For dynamic simulation studies and stability studies, CTU stated presently there are convergence issues being faced in dynamic simulation for 2027 scenario for which consultation with IIT Mumbai is also under progress. In view of that, Dynamic simulation studies for planning scenario may take some time.

In 30th CMETS-NR meeting, CTU requested that Grid-India may also share present time frame converged dynamic file which will help to resolve convergence issues in dynamics file for 2027 timeframe w.r.t Generator models (conventional/RE) used by Grid-India. In the CMETS-NR meeting Grid-India stated that they are also facing convergence issues for present timeframe file and prepared dynamics file with some assumptions. They also shared some dynamic data with CTU in the past but only dynamic data will not help to resolve convergence issue. However for preparation for planning stage dynamic file, CTU and Grid-India can work together to prepare load flow first w.r.t dynamics file and subsequent Dynamic data file (Dyr) shall be prepared with help of models available with Grid-India.

CTU also stated that considering the severe convergence issues in dynamics file, it will be converged on truncated network specific to RE pockets in Rajasthan rather than to converge the file on All India level. NRLDC stated that for transient studies, truncated network modelling will be sufficient.

Further, in the meeting CTU stated that to resolve the issue of higher angular separation (>20 degree) under N-1/N-1-1 contingency of 765kV Bikaner-Moga D/c line, suitable strengthening scheme is already under planning. As part of above scheme LILO of 765kV Bikaner -Moga line is envisaged which will reduce the angular separation (<30 degree). The above scheme will be taken up in Jul'24 meeting.

Regarding observation for Inertia response of units, NRLDC stated that it may be checked in dynamics file once prepared for critical contingencies i.e. outage of large generation say 5-10GW in RE pockets. In the above condition, inertial response of generation units, Grid frequency variation, df/dt, synchronous condenser response may be analyzed. Same was agreed.

- O. Wherever long lines are getting planned like in this case 765 kV Sirohi-Mandsaur (~320 Km), the range of line length may be provided as it is possible that route length may increase during commissioning phase.

CTU Reply: BEE length of the 765 kV Sirohi-Mandsaur D/c Line is about 265kms (line length would be about 320kms considering 20% on BEE length) considering Substation location for Sirohi and Mandsaur which is already finalized with boundary limits.

Line Lengths of all the transmission lines in proposed scheme will be reviewed in Gati Shakti portal in due course time and in case of any change, impact of same will be checked in studies w.r.t angular separation and voltage limits. Accordingly, requirement of change in reactive compensation will also be reviewed, if needed.

- P.** Intra-state network wherever connected or proposed to be connected may be specified. All such long HVDCs and AC lines will impact Rajasthan ATC/TTC.

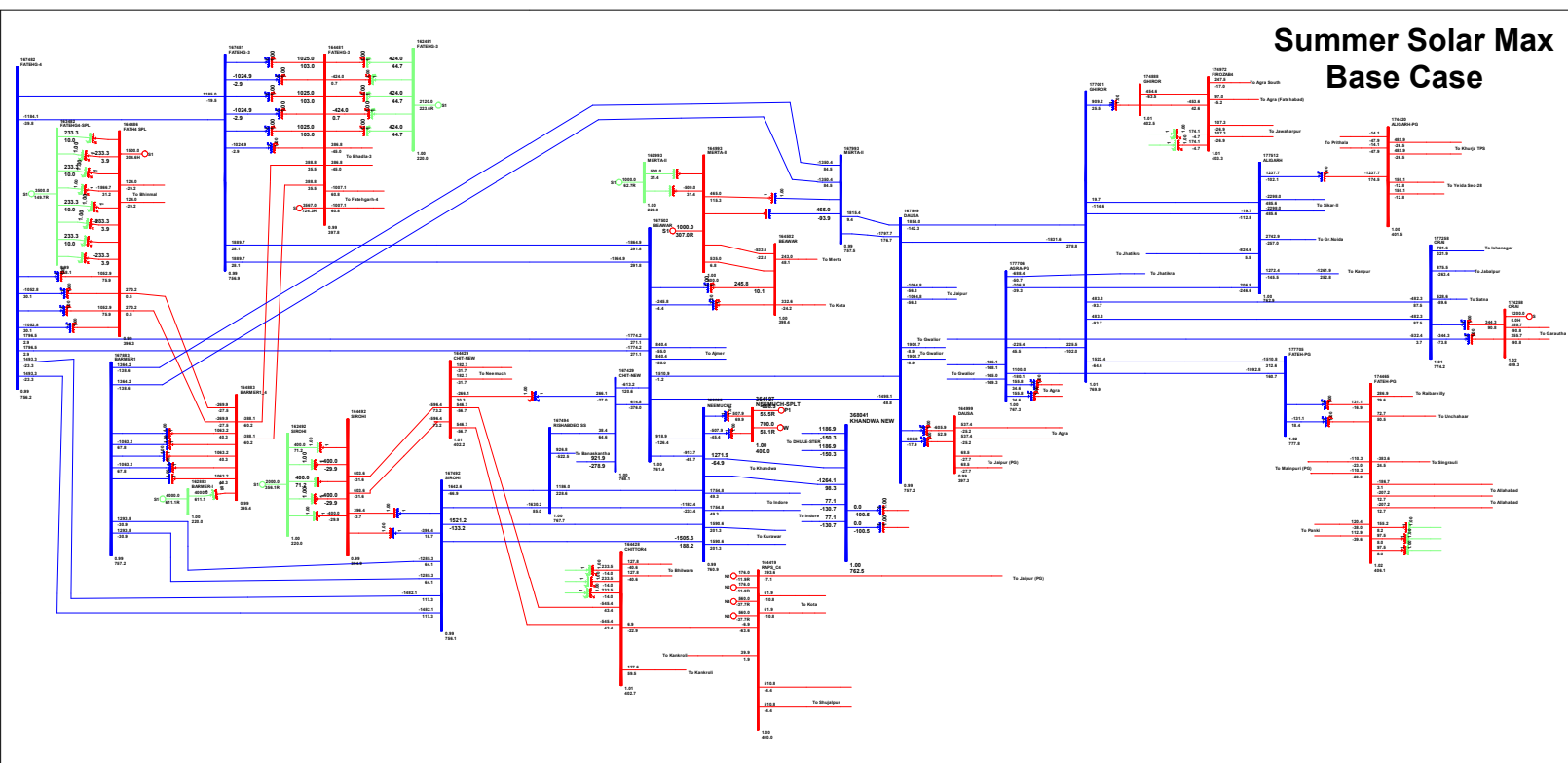
CTU Reply: No interconnection is being planned from intra state network of RVPN as part of proposed scheme or earlier agreed transmission schemes i.e. Rajasthan REZ Ph-IV/V for RE evacuation

- Q.** Name of New Substations – It is requested to assign unique name for envisaged RE pooling stations. As several pooling substations with identical names but different numerical suffixes have been planned in the same district along with multiple interconnections, there is a chance of miscommunication during real-time operation among multiple constituents. For clarity of operation and ensuring that names of different substations are easily distinguished from each other, it is suggested that after finalization of the exact location, the ISTS substations are uniquely named as per the geographical name of the nearest location like village or taluk.

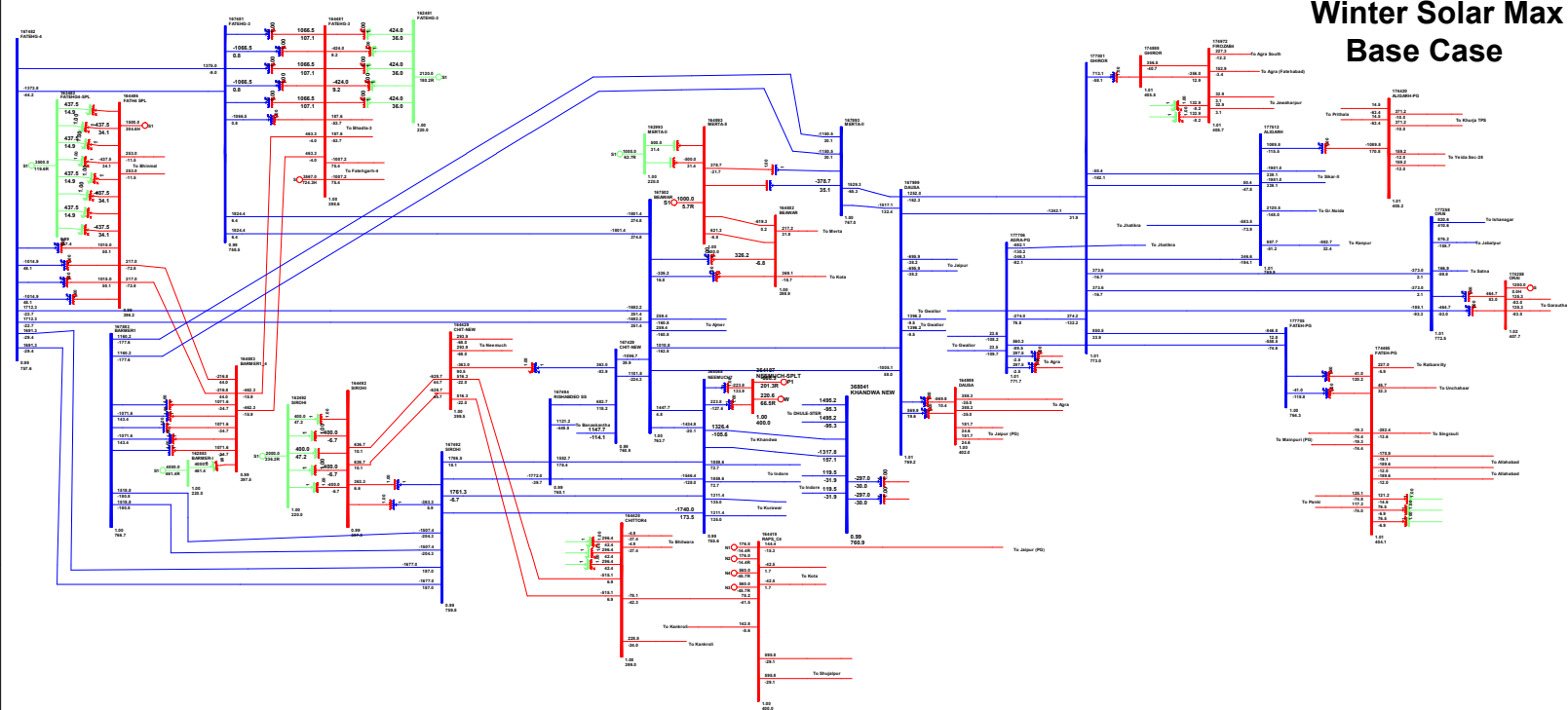
CTU reply: The issue has been highlighted many a times at OCC/RPC level. CTU vide mail 17.06.24 again requested to TSPs that being owner of the substations, renaming of Pooling Stations may be carried out. In the meeting, CTU stated that naming convention should be unique for every pooling station in RE complex in combination of existing name along with village/taluk name. POWERGRID stated that they will revert on the same with resolution.

Exhibit-I

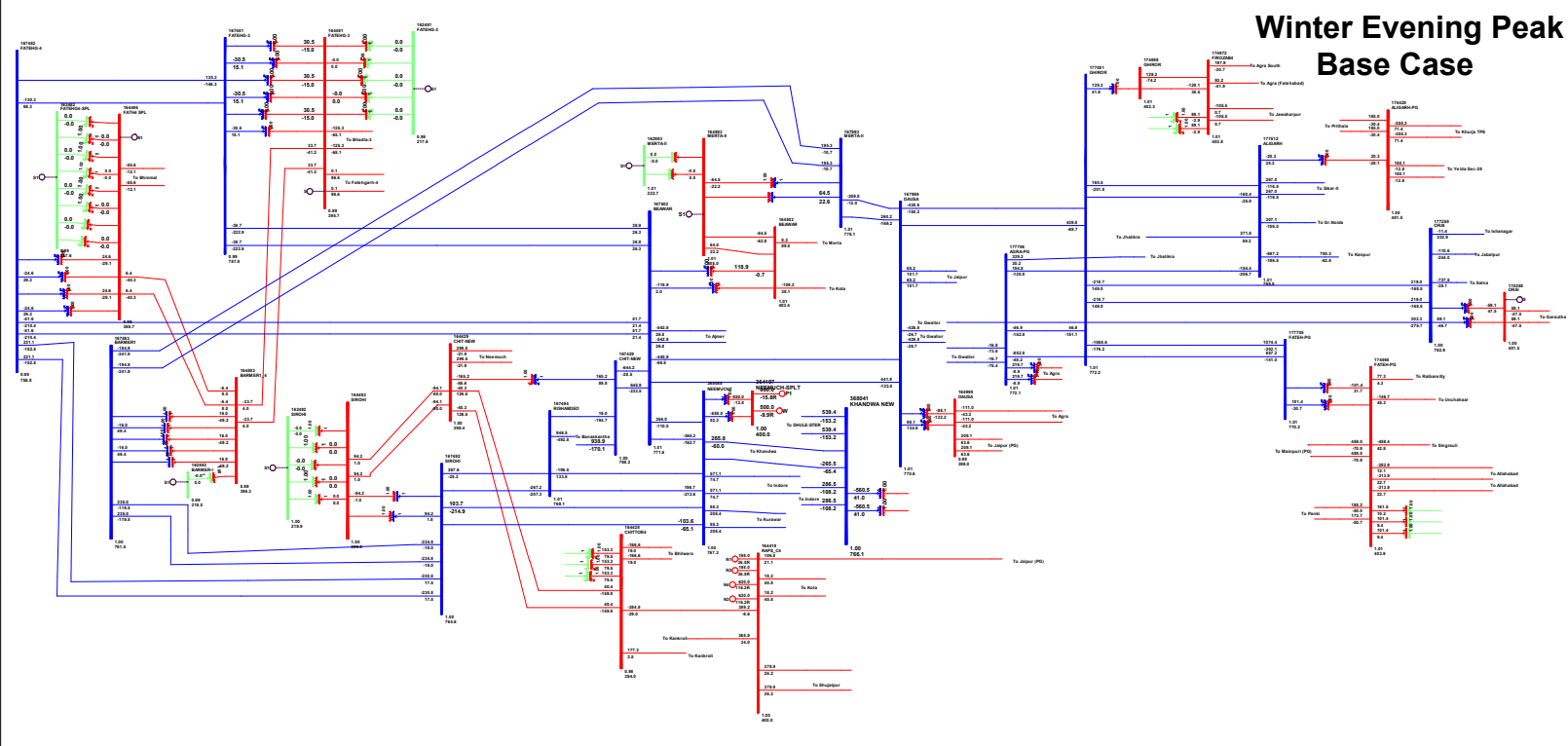
Summer Solar Max Base Case



Winter Solar Max Base Case



Winter Evening Peak Base Case



Winter Night off Peak Base Case

